



**UNIVERSITY
OF ICELAND**

Meteorological Conditions and Rural Injury Accidents in Iceland

Analysis of Accident Frequency and Traffic, 2007–2025

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M.Sc. thesis
in Software Engineering

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30 ECTS thesis submitted in partial fulfilment of a
Master of Science degree in Software Engineering

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Abstract

This study examines whether rural injury accidents in Iceland occur disproportionately during strong wind. Accident records from 2007–2025 were linked to nearby weather observations. The primary analysis compared observed accident counts with those expected from the local frequency of each weather condition. Two further analyses examined the role of traffic: an approximate adjustment for changes in traffic during adverse weather, and a separate comparison of accident rates using daily traffic counts and estimated distance travelled. Mean wind was the primary weather measure; gust and temperature provided complementary comparisons.

Strong wind was over-represented at accident times. At mean wind speeds of at least 20 m/s, the observed-to-expected accident ratio (O/E) was 2.15. Counters showed lower traffic in strong wind. Accounting approximately for this reduction strengthened rather than removed this pattern. The separate traffic-based rate analysis pointed in the same direction, although its highest-wind intervals contained few accidents. Gust showed a similar pattern, while temperature associations were less consistent across methods.

The findings support an association between strong wind and rural injury accidents, not a causal effect. Weather observations cannot fully describe conditions on the road, and the traffic analyses rely on selective counter coverage and estimated exposure within each day. Hourly traffic data with broader geographic coverage would be needed for a direct national estimate of accident risk per distance travelled.

Ágrip

Rannsóknin fjallar um tengsl veðurs og umferðarslysa með meiðslum í dreifbýli á Íslandi á árunum 2007–2025. Kannað var hvort slys væru hlutfallslega algengari í miklum vindi en við önnur veðurskilyrði. Slysaskrár voru tengdar veðurmælingum frá nálægum stöðvum og fjöldi slysa borinn saman við þann fjölda sem vænta mætti miðað við hversu oft viðkomandi veður kom fyrir á hverjum stað. Einnig var tekið tillit til þess að umferð getur minnkað í slæmu veðri. Annars vegar var samanburðurinn leiðréttur með hliðsjón af gögnum um daglega umferð og hins vegar var slysatíðni metin miðað við áætlaðan akstur á vegum þar sem umferðarteljarar voru til staðar.

Slys reyndust hlutfallslega algengari í miklum vindi. Við meðalvindhraða 20 m/s eða meiri var hlutfall skráðs og væntanlegs fjölda slysa 2,15. Umferðarmælingar sýndu minni umferð í sterkum vindi og leiðrétting vegna þess styrkti þetta mynstur. Samanburður við áætlaðan akstur benti í sömu átt, en fá slys lágu að baki niðurstöðum fyrir mesta vindinn. Svipuð tengsl komu fram fyrir vindhviður. Niðurstöður fyrir hitastig voru óreglulegri og háðari greiningaraðferð.

Niðurstöðurnar sýna tengsl en staðfesta ekki að vindur hafi valdið slysum. Veður á mælistöð lýsir ekki alltaf aðstæðum á vegi og umferðarteljarar ná hvorki til allra vega né allra tímabila. Þar að auki lágu ekki fyrir umferðartölur eftir klukkustundum. Víðtækari og nákvæmari umferðarmælingar þarf því til að meta slysaáhættu á hvern ekinn kílómetra fyrir landið allt.

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Abbreviations

CI	Confidence interval.
O/E	Observed-to-expected accident ratio.
OR	Odds ratio.
RR	Rate ratio.
VKT	Vehicle-kilometres travelled.

1 Introduction

1.1 Context and Motivation

Weather affects both driving conditions and the decision to travel. Strong wind can make a vehicle harder to control, low temperature can coincide with slippery roads, and darkness can reduce visibility. These conditions are relevant to Icelandic roads, where terrain can produce substantial local variation in wind (Snæbjörnsson et al., 2007; Pu et al., 2018). Precipitation and fog are also relevant road-weather hazards, but they are context for this study and are not analysed here.

Previous research provides two reasons to examine wind. Engineering studies describe mechanisms by which crosswind can destabilise vehicles, while observational studies compare accident occurrence with the frequency of windy conditions (Baker, 1986, 1994; Edwards, 1994). These are complementary forms of evidence: a physical mechanism does not establish how often accidents occur across a road network. Reviews also find less consistent road-safety evidence for wind than for precipitation (Theofilatos and Yannis, 2014). Local evidence is therefore needed to determine whether the same pattern is present on Icelandic roads.

In Iceland, wind-related road safety has been studied using vehicle-reliability analyses and operational vehicle-wind models (Snæbjörnsson et al., 2007; Hrafnkelsdóttir, 2020). This thesis addresses a different question: whether rural injury accidents are over-represented under particular weather conditions relative to how often those conditions occur locally. It then examines how the interpretation changes when available traffic information is introduced. The study therefore links a long Icelandic accident series with local weather frequency and two complementary traffic-exposure approaches, while treating station observations as proxies for road conditions rather than direct measurements of the forces acting on a vehicle.

The study covers rural injury accidents in Iceland from 2007–2025. Accident times and locations are linked to nearby 10-minute weather observations. Mean wind is the primary weather measure; wind gust and temperature are secondary measures, while season, time of day and daylight provide additional context.

Accident counts alone do not measure accident risk. Severe weather may increase the hazard faced by each vehicle while also reducing the number of vehicles on the road. Weather frequency and traffic volume are therefore considered separately (Andrey et al., 2003; Lord and Mannering, 2010; Washington et al., 2020).

1.2 Aim and Research Questions

The primary research question is: *How does rural injury accident occurrence vary with weather, particularly strong wind, after accounting as far as possible for local weather*

frequency and traffic exposure?

The analysis proceeds in three stages:

1. **Weather-frequency baseline (Q1).** Compare observed accidents with the number expected from local weather frequency, by parameter and season. This measures over-representation relative to weather time, not risk per vehicle.
2. **Approximate traffic correction (Q2).** Establish how traffic changes with weather, then use the observed counter response to reweight the baseline comparison. This asks whether lower traffic explains the wind pattern.
3. **Counter-based accident rates (Q3).** Relate accident counts to estimated vehicle-kilometres using daily traffic, rural road length and monthly weather frequencies. Compare with allocation using the actual day's weather.

Q1–Q3 identify analysis stages addressing this question, not repeated estimates of one parameter. The baseline and approximate correction cover 2007–2025; the rate analysis uses a smaller counter-linked sample from 2019–2024. Matched-time and allocated daily-counter models provide secondary robustness checks. Minor injury and serious or fatal injury form disjoint components of all injury. The study estimates associations, not causal effects or a complete national accident rate per vehicle-kilometre.

1.3 Thesis Outline

Chapter 2 gives the background for the weather and traffic comparisons. Chapter 3 describes the data, data preparation, and statistical methods. Chapter 4 presents the three main analyses in research-question order, followed by supporting checks. Chapter 5 discusses their interpretation and limitations, and Chapter 6 gives the conclusions. The appendix contains additional checks that support interpretation of the main results.

2 Background

2.1 Wind and Rural Road Accidents

Crosswind can produce overturning, lateral sideslip, or excessive yaw. The critical condition depends on vehicle geometry and mass, vehicle speed, road friction, wind direction, and driver response rather than wind speed alone (Baker, 1986, 1994). High-sided vehicles are particularly exposed, but the physical mechanisms also provide a reason to examine loss-of-control accidents more generally. These engineering studies establish physical plausibility and vehicle-specific failure mechanisms; they do not by themselves establish a population-level association between wind and accident occurrence.

This problem is directly relevant in Iceland, where terrain can accelerate and redirect near-surface wind over short distances. An Icelandic reliability study by Snæbjörnsson et al. (2007) modelled accident probability as a combined function of wind velocity and direction, road camber and friction, and vehicle speed. A station measures wind at one point, whereas vehicle stability also depends on the wind's direction relative to the road and on local road conditions. Terrain between a station and a road can alter the wind reaching the vehicle. Station observations therefore remain exposure proxies, and a shorter matching distance does not guarantee identical conditions. Mobile road measurement research similarly shows that roadside terrain creates marked spatial variation that a sparse fixed-station network cannot fully resolve (Pu et al., 2018).

Vegagerðin has developed a vehicle-wind model to support decisions about hazardous conditions and road closures, while also noting the limited experimental evidence available for passenger cars (Hrafnkelsdóttir, 2020). The present thesis does not test that operational model; it examines population-level accident occurrence using a separate data design.

Empirical work has compared the share of accidents occurring in high wind with the share of time that high wind was observed, an approach closely related to the observed/expected comparison used here (Edwards, 1994). Studies linking crashes to nearby weather stations likewise show that station observations can be informative while retaining measurement error (Young and Liesman, 2007). More broadly, Theofilatos and Yannis (2014) report less consistent road-safety evidence for wind than for precipitation.

2.2 Temperature, Daylight, and Time of Day

Temperature, daylight, and time of day add context that wind alone cannot provide. They affect road-surface conditions, visibility, travel patterns, and the composition of road users, but accident counts in these categories are not rates unless the corresponding amount of travel is known. Temperature effects can also be non-linear and season-specific: a Seoul time-series

study found its strongest cold-weather association below freezing in winter, while an hourly case-crossover study in Shenzhen found a J-shaped temperature relationship (Lee et al., 2014; Zhan et al., 2020). Temperature is therefore compared with its local seasonal frequency rather than treated as having one linear effect.

Darkness presents the same denominator problem. A method developed using Norwegian, Swedish, and Dutch data compared daylight and darkness within clock time and season, avoiding the interpretation of raw night-time counts as risk (Johansson et al., 2009). Here, raw hour, season, and daylight counts describe the sample. Matched daylight and adjusted severity models are used for numerical comparisons.

2.3 Traffic and Accident Rates

Weather can change the amount of traffic as well as the hazard. Research using automatic traffic recorders shows that extreme weather variables, including wind, can reduce or redistribute hourly traffic volume (Sathiaraj et al., 2018). Accident counts during strong wind can remain low even if risk to each travelling vehicle is higher. High-resolution crash studies illustrate the value of hourly weather data rather than daily or monthly averages (Xing et al., 2019). The analyses distinguish three quantities: accident occurrence relative to weather frequency, rates based on estimated vehicle-kilometres, and changes in observed daily traffic. Studies with real-time traffic, weather, and road geometry show what a more complete analysis could achieve when such data are available (Ahmed et al., 2012).

3 Data and Methods

3.1 Data

3.1.1 Source Data

The data were obtained from four government institutions: the Icelandic Transport Authority (*Samgöngustofa*; ITA), the Icelandic Road and Coastal Administration (*Vegagerðin*; IRCA), the Icelandic Meteorological Office (*Veðurstofa Íslands*; IMO), and Statistics Iceland (*Hagstofa Íslands*); see Table 3.1.

Table 3.1: Source datasets and coverage.

Data	Provider	Contents	Coverage	Reference
Accidents	ITA	Accident events	126,607 events, 2007–2025	ITA (2026)
Weather	IMO	10-minute observations	232,459,562 observations; 338 stations, 2007–2025	IMO (2026)
Traffic/roads	IRCA	Seasonal averages, daily counts and road geometry	1,226 mapped road sections; 33,757 section-year records of seasonal averages, 2000–2025; 1,033,659 daily channel records, 2019–2024	IRCA (2026)
Urban areas	Statistics Iceland	Urban boundaries	97 polygons, 2020–2024 release	Statistics Iceland (2026)

Records from the year 2025 are included in the accident, injury, vehicle, road, seasonal-traffic and weather datasets under the same selection and quality criteria as records from earlier years. Daily counter data are available through 2024. Table 3.2 defines the weather and traffic quantities used in the subsequent comparisons.

Table 3.2: Definitions used throughout the thesis.

Term	Definition	Use in this thesis
f	10-minute mean wind speed, m/s	Main weather measure, matched within five minutes of the accident time.
fg	Wind gust reported in the same 10-minute observation, m/s	Additional weather measure; not a daily maximum.
t	Air temperature reported with each 10-minute observation, °C	Additional weather measure matched independently.
O/E	Observed accidents divided by accidents expected from local weather frequency	Main relative-occurrence result; traffic is not included.

3.1.2 Accident Data

The accident dataset contains 126,607 records from 2007–2025 (ITA, 2026). One record lacks valid temporal or spatial information, leaving 126,606 valid records.

Rural status is derived from accident coordinates and Statistics Iceland urban-locality polygons (Statistics Iceland, 2026). Points inside an urban polygon are classified as Urban, and valid points outside all polygons as Rural. A defined Vestmannaeyjar area is also classified as Urban. The same 2020–2024 polygon dataset is applied to the full 2007–2025 accident period. Changes in urban boundaries before 2020 are not represented.

Applying this rural classification yields 24,005 accidents, of which 6,414 involve injury across the rural road network (Figure 3.1). Of these, 6,259 have a qualifying primary weather match within 20 km and five minutes.

Accident severity is recorded in `meidsli`: code 1 denotes a fatal accident, code 2 a serious accident, code 3 a minor-injury accident, and code 4 an accident without injury. The main outcome is $\text{meidsli} \leq 3$. The serious or fatal comparison uses $\text{meidsli} \leq 2$. The source data also contain 17,591 rural damage-only accidents; these are outside the injury outcome.

For the reader-facing results, injury accidents are divided into two disjoint groups: minor injury ($\text{meidsli}=3$) and serious or fatal injury ($\text{meidsli}=1$ or 2). The O/E calculation for these groups is described in Section 3.3.1.

Table 3.3 shows one cleaned rural injury accident record before weather matching.

Table 3.3: Example of a cleaned rural injury accident record. The table illustrates accident-level information retained before weather matching. The source identifier alone is masked; time and coordinates are reproduced as recorded.

Field	Example value
Example identifier	A (source identifier omitted)
Date	2007-01-03
Recorded time	18:08:00
Registered road section	1-g8
Latitude / longitude (degrees)	64.57709888007743, -21.879289941375845
Area classification	Rural
Injury category	Minor injury
Involved vehicles	1

3.1.3 Weather Data

The weather dataset contains 10-minute observations from Icelandic weather stations. The variables used here are 10-minute mean wind speed (f), reported wind gust (fg), and temperature (t). For each accident, the gust is taken from the same matched 10-minute observation as mean wind; it is not the maximum across the accident day. Mean wind speed remains the main weather measure. A wind observation is used when

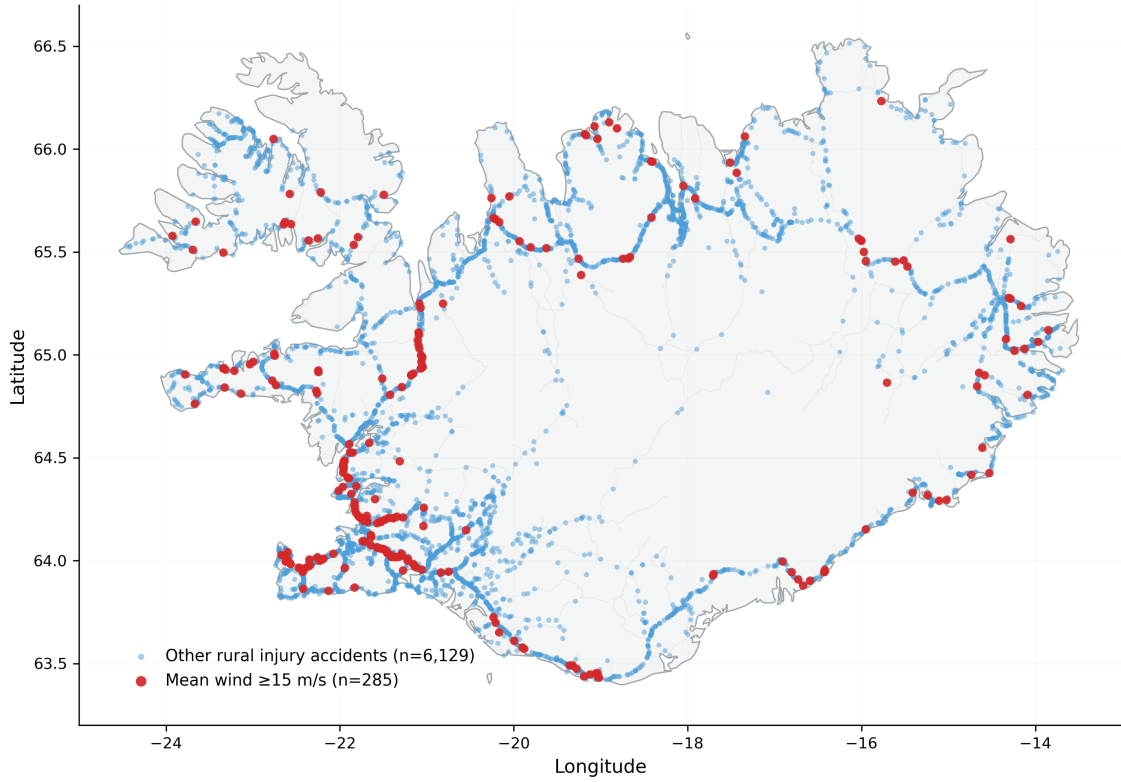


Figure 3.1: Locations of the 6,414 rural injury accidents. Accidents matched to mean wind of at least 15 m/s are highlighted. The map shows the spatial coverage of the study. Accidents follow the main rural road network; strong-wind matches are geographically dispersed.

$$0 \leq f < 45, \quad 0 \leq fg < 65, \quad fg + 0.5 \geq f.$$

Rows missing either f or fg are excluded. Negative values, values at or above either upper limit, and internally inconsistent values where $fg = 0$ but $f > 0$ are excluded. These quality exclusions are summarised in Table 3.5. A value of $f = 0$ alone is kept. Only uninterrupted all-zero runs where $f = fg = 0$ for at least two hours are excluded as potentially frozen sensors. Missing temperature does not affect whether a wind observation is used.

Each accident is matched to the geographically nearest station with a quality-controlled wind observation at one of the two adjacent timestamps on the 10-minute observation grid. The selected observation is therefore no more than five minutes from the recorded accident time. Because matching is based on absolute time difference, the adjacent observation can fall on the neighbouring calendar day for accidents occurring within five minutes of midnight. The maximum matching distance is 20 km. The matching dataset includes all 6,414 accidents and records whether each accident meets the weather-match criterion.

Temperature is matched independently within the same 20 km and five-minute limits. The nearest valid value between -30 and 30 °C may therefore come from a different station when the wind station lacks temperature. In the final analysis dataset, however, all 6,259 wind and temperature matches use the same station and observation time.

The IMO supplied observations of time, station, temperature, mean wind, and gust. The same quality-controlled weather data are used for accident matching, background frequencies, matched control times, and traffic analyses. No temperature value is imputed.

Table 3.4 shows the coverage gained as the matching radius increases. A distant station is a less desirable proxy for road conditions, but there is no physically exact cut-off. The 20 km threshold was selected as a practical compromise between high coverage and avoiding very distant stations; it was not determined by an optimisation procedure.

Table 3.4: Weather-match coverage at alternative station-distance thresholds. The denominator is 6,414 rural injury accidents, 2007–2025; all matches require valid mean wind and gust within five minutes.

Maximum distance (km)	Matched accidents	Matched (%)
5	3,060	47.7%
10	5,098	79.5%
20	6,259	97.6%
30	6,402	99.8%

Table 3.5: Overview of cleaning of wind measurement records, 2007–2025. Cleaning-rule shares use all supplied observations. Every supplied station-year contains at least one wind measurement.

Category	Records	Share
All supplied station-time observations	232,459,562	100.00%
Missing f or fg	261,519	0.11%
$f < 0$, $fg < 0$, $f \geq 45$ or $fg \geq 65$ m/s	8,766	0.00%
$fg = 0 < f$ or $fg + 0.5 < f$ (m/s)	257,245	0.11%
All-zero runs ≥ 2 hr	1,473,082	0.63%
Wind observations retained	230,458,950	99.14%

The largest excluded category is uninterrupted all-zero runs, which is reported separately because it reflects the rule for possible instrument stoppages rather than a missing observation. The audit reports each rule separately and uses all supplied weather observations from 2007–2025. For each accident’s station and season, the O/E expected count uses all quality-controlled background observations in that station–season group, not only observations recorded on accident days.

3.1.4 Traffic Data

Daily counters provide the traffic information used in the main analysis. Seasonal road-section averages are also available for 2000–2025 (IRCA, 2026). In principle, summer and winter averages and a residual spring/autumn average could be spread over their respective periods to approximate exposure across the full study period. This is not the main approach used here; these averages are not used in the main analysis.

Throughout the analyses, the four study seasons are aligned with these traffic periods: winter is December–March, spring is April–May, summer is June–September, and autumn is October–November. These are study-specific traffic-aligned seasons rather than conventional three-month meteorological seasons. Using June–September as summer keeps the weather

comparison aligned with the Icelandic road authority’s summer traffic period; it is an operational definition, not a claim that weather is uniform across those months. The IMO also uses June–September when summarising the Icelandic summer (Veðurstofa Íslands, 2025). Daily traffic counts extracted from Vegagerðin reports cover 2019–2024 and provide observed totals at selected counter sites. They underpin the traffic response and the counter-linked exposure calculation, but do not provide hourly traffic at accident locations. More than one counter may occur on a road section, and counter positions are estimated from the registered road geometry. These data remain separate from the primary O/E calculation.

3.1.5 Selection Summary

Table 3.6 summarises the main data reductions. Selection for the study area and injury outcome is distinguished from invalid or missing measurements: an excluded record is not necessarily an erroneous record. The traffic-based rate analysis starts with the 1,863 rural injury accidents in 2019–2024; the broader weather sample covers 2007–2025. The first four rows follow the accident selection sequentially. The last three rows describe separate selections for counter-linked accidents, weather observations and counter-days.

Table 3.6: Overview of primary data selections for accidents, weather and traffic counts. The first four rows are sequential; the last three rows start from separate populations. Each removal percentage uses the count entering that row.

Selection	Entering	Removed at step (%)	Retained	Reason
Valid accident records	126,607	< 0.01%	126,606	Invalid time or coordinates.
Rural accidents	126,606	81.04%	24,005	Inside the urban study boundary.
Rural injury accidents	24,005	73.28%	6,414	No recorded injury.
Accidents for weather-only analysis	6,414	2.42%	6,259	No qualifying weather within 20 km and five minutes.
Accidents for VKT analysis	1,863	62.75%	694	Separate 2019–2024 sample: counter linkage, daytime, weather and traffic eligibility.
Wind observations	232,459,562	0.86%	230,458,950	Wind exclusions in the cleaning overview.
Daily counter-days	774,274	1.43%	763,171	No usable daytime wind summary.

3.1.6 Descriptive Accident Patterns

Figure 3.2 shows when the selected accidents occurred and the conditions recorded with them. The blue and red groups are the same separate severity groups used in the O/E figures. These are descriptive counts: more accidents in an hour or season may reflect more traffic, and the panels do not estimate accident risk. Vehicle counts and accident-type composition are shown in Appendix A.1. Serious or fatal injury accidents are more numerous in summer than winter, while the minor-injury counts are similar. The longer summer and winter periods and differences in travel must be considered when reading these counts.

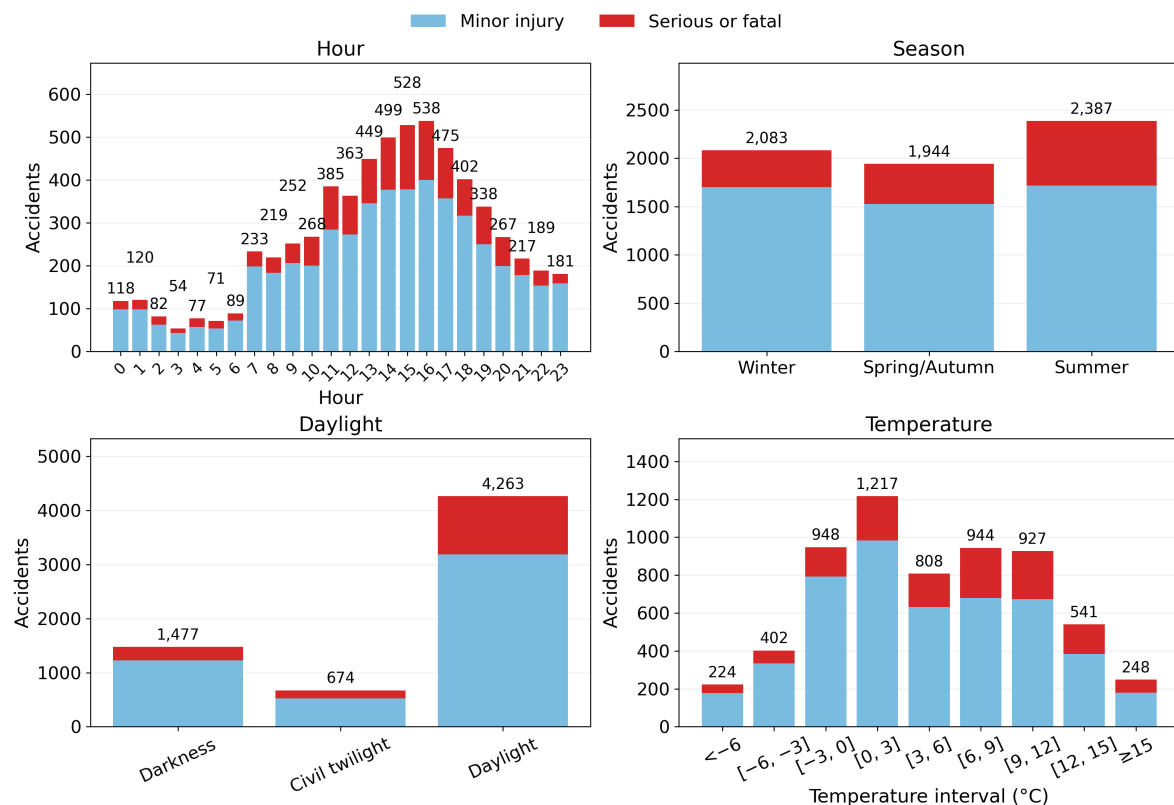


Figure 3.2: Distribution of rural injury accidents by hour, study season, estimated daylight and matched temperature. Blue shows minor injury (code 3), and red shows serious or fatal injury (codes 1–2). Spring and autumn are combined only in this descriptive figure. Hour, season and daylight use 6,414 accidents; temperature uses 6,259 qualifying matches. Temperature brackets denote displayed interval endpoints; bins include the lower endpoint and exclude the upper. These are counts without traffic adjustment.

3.2 Data Preparation and Reproducibility

Table 3.7 summarises the path from source data to the three main analyses. Accident, weather and traffic records first undergo quality checks and selection. Qualifying accidents are matched to nearby weather observations, and local weather frequencies form the denominator for the weather-only analysis. The correction reweights this comparison using the traffic response estimated from the 2019–2024 counters. The traffic-based rate analysis uses the counter-linked 2019–2024 sample to estimate vehicle-kilometre exposure from observed daily

traffic, road length and pooled monthly weather frequencies. Matched-time and additional traffic models are retained as supporting checks. This separation is important because the analyses use different exposure assumptions and eligible samples.

All cleaning, matching, analysis, and thesis-table generation are scripted. Generated tables are written from analysis outputs, and validation checks cover key sample counts, matching rules, and exposure conservation. This reduces manual transcription between the analysis and the thesis while keeping the main transformations auditable.

Table 3.7: Data preparation and analysis workflow.

Stage	Main data	Output	Role in thesis
Data preparation	Accident, weather, road and traffic data	Quality-controlled datasets	Defines the study data.
Accident selection	Accident data and urban boundaries	Rural injury accidents	Defines the study population.
Weather matching	Accidents, weather observations and stations	Accident-time weather and local frequencies	Common input to the weather analyses.
Weather-only O/E	Accident weather and local frequencies	Observed and expected counts	Primary analysis.
Approximate traffic-corrected O/E	Primary O/E and 2019–2024 counter response	Traffic-corrected expected counts	Approximate traffic-exposure sensitivity.
VKT analysis	Recorded counter-days, rural lengths and monthly weather frequencies	Accidents per million vehicle-km	Main counter-linked traffic analysis.
Supporting analyses	Matched control times and alternative traffic panels	Matched odds ratios and alternative rate estimates	Robustness checks.

3.3 Analysis

3.3.1 Weather-only Observed / Expected Analysis

The observed-to-expected ratio (O/E) compares the number of accidents observed in each weather interval with the number expected from local weather frequency.

Accident-time mean wind speed f is the main weather measure and is divided into 5 m/s intervals. It is the 10-minute mean wind observation matched within five minutes of the accident time, not an instantaneous measurement. Wind gust f_g from the same observation and independently matched temperature are additional weather measures. For each measure, accidents are compared with the distribution of that weather measure at the matched station in the same study season, pooled across all study years. Expected counts are therefore based on how often each condition occurred locally rather than on a national weather distribution. For example, if 5% of the relevant station–season observations fall in one wind interval, approximately 5% of that group’s accidents would be expected there under weather-frequency-proportional occurrence.

Let g identify a weather station and study season. Let A_{jg} be the number of accidents in weather interval j and group g , and let $A_g = \sum_j A_{jg}$. From all quality-controlled 10-minute

observations, let T_{jg} be the number in interval j and T_g the total in group g . The local background frequency is

$$p_{jg} = \frac{T_{jg}}{T_g}.$$

If accident timing followed local wind frequency, the expected number in the interval would be $A_g p_{jg}$. The pooled observed/expected ratio is

$$R_j = \frac{\sum_g A_{jg}}{\sum_g A_g p_{jg}}.$$

At mean wind of at least 20 m/s, 56 minor-injury accidents were observed compared with 25.18 expected, giving O/E 2.22. This illustrates the calculation without attributing any individual accident to wind.

A value of one means that the accident share equals the local time share of the weather interval. A value above one means that accidents are over-represented in that interval. Standardisation within station and season prevents locations or seasons with different weather distributions from dominating the result.

The figures show the calculation separately for minor-injury accidents ($mei\delta sli = 3$) and serious or fatal accidents ($mei\delta sli \leq 2$). These groups do not overlap. Each group's expected count uses its own accident totals within station and season. The combined all-injury O/E is also retained for comparison with the other analyses; it is calculated from summed observed and expected counts, not by averaging the two group ratios. The weather-only analysis is reported as a descriptive standardised ratio rather than a fitted inferential model, so observed and expected counts are shown without model-based confidence intervals. Wind gust and temperature are handled in the same way as mean wind. Temperature is matched independently, and its intervals are below -6 , -6 to -3 , -3 to 0 , 0 to 3 , 3 to 6 , 6 to 9 , 9 to 12 , and at least 12°C . In all temperature figures, square brackets show the interval endpoints as a display convention: each finite bin includes its lower endpoint and excludes its upper endpoint. The O/E upper interval is combined at 12°C to keep the seasonal panels readable. The matched-time models retain separate 12 – 15 and at-least- 15°C intervals. Estimates across those different upper intervals are not treated as direct numerical comparisons.

3.3.2 Use of Traffic Data

The primary O/E denominator is the local frequency of wind conditions rather than traffic volume. Complete traffic data are unavailable at the required spatial and temporal resolution, so the weather-only analysis does not account for changes in travel volume during strong wind. If traffic is lower in strong wind while other factors remain unchanged, the accident-rate contrast per vehicle-kilometre would be larger than the weather-frequency O/E contrast.

Traffic enters the main analysis in two ways. The approximate correction estimates the relative daily traffic response across weather intervals from the 2019–2024 counters and uses that response to reweight weather-only denominator. The traffic-based rate analysis constructs estimated vehicle-kilometre exposure from observed daily traffic and rural road length in a smaller counter-linked sample. Neither analysis observes hourly traffic. An

allocated daily-counter model is retained as a supporting check under an alternative exposure construction.

3.3.3 Approximate Traffic-Corrected O/E Analysis

The approximate correction uses the observed day-to-day traffic response at counter sections during 2019–2024. Among positive-count days, the mean count for the same counter section, year, month, and weekday defines expected daily traffic. Each day’s observed 24-hour count and expected count are allocated to weather intervals in proportion to their valid 07:00–24:00 weather minutes on that date. For interval j , r_j is the sum of allocated observed traffic divided by the sum of allocated expected traffic. Zero-count days are excluded from this response calculation.

Using the same station–season group g and weather interval j as in the weather-only analysis, the traffic-weighted weather frequency is

$$p_{jg}^* = \frac{p_{jg} r_j}{\sum_h p_{hg} r_h}.$$

Expected counts are then recalculated as $A_g p_{jg}^*$. Accident bins use weather at the recorded accident time, whereas r_j is based on a 24-hour traffic total repeated across the day’s daytime weather composition. Daily counters do not observe hourly traffic. The 2019–2024 response is also extrapolated to the 2007–2025 primary population. The result is therefore a model-based sensitivity analysis, not a directly observed traffic-adjusted O/E.

The period sensitivity repeats the original comparison separately for 2007–2018 and 2019–2024, restricting both accidents and background weather to the stated years. The latter period is also recalculated with the counter traffic response. These All year comparisons show what changes when the accident and weather period aligns with the traffic data. Minor and serious or fatal injuries remain disjoint and use their own expected counts. The existing counter-era correction includes temperature as well as wind and gust; it does not extend the full-period correction to temperature.

3.3.4 VKT Analysis

The main traffic rate is restricted to 2019–2024, when daily counter data are available. Eligible accidents must be rural injury accidents occurring between 07:00 and 24:00, have a counter section for the registered road and year, be successfully located on that section, have usable accident-time wind and gust within 20 km and five minutes, and have a traffic record on the accident date. For this analysis, the 20 km weather distance is measured from the counter-section location rather than from the accident coordinate. The same five-minute time rule applies as in the weather-only analysis and can select the adjacent observation across midnight. The denominator is broader than the accident set: it uses every eligible recorded counter-section day with a nonnegative traffic count and rural road length, including days without an accident. For counter section j , date d , and weather interval b , estimated exposure, in vehicle-kilometres travelled (VKT), is

$$\hat{V}_{jdb} = C_{jd} L_j p_{j,m(d),b},$$

where C_{jd} is the observed 24-hour traffic count, L_j is rural section length under that year's definition, and $p_{j,m(d),b}$ is the 07:00–24:00 weather frequency for that calendar month at the station assigned to counter section j . The weather frequencies are pooled across 2007–2025 to provide stable station–month proportions, while both traffic and accident eligibility remain restricted to 2019–2024. The full daily count is allocated to the 07:00–24:00 window and the proportions sum to one, so exposure is conserved. Accidents are classified separately by observed event-time weather within 20 km and five minutes. The reported rate is

$$R_b = \frac{O_b}{\sum_j \sum_d \hat{V}_{jdb}} 10^6.$$

This construction uses actual day-to-day traffic variation and rural road length, but allocates it with a typical monthly weather distribution rather than observed hourly traffic. Daily traffic still varies from day to day, but one storm day's weather does not determine that day's entire allocation among weather bins. The typical distribution for the station and calendar month provides a more stable allocation. This is an exposure assumption, not evidence that the monthly method measures actual traffic within each weather interval.

For the injury-component figures, the exposure denominator $\hat{V}_b = \sum_j \sum_d \hat{V}_{jdb}$ is shared:

$$R_b^{\text{minor}} = \frac{O_b^{\text{minor}}}{\hat{V}_b} 10^6, \quad R_b^{\text{serious/fatal}} = \frac{O_b^{\text{serious/fatal}}}{\hat{V}_b} 10^6.$$

Minor injury means code 3; serious or fatal injury means codes 1–2. Their counts are disjoint, so their rates sum to the all-injury rate. Blue forms the bottom of each stack and red the top. This is a descriptive decomposition of the accident numerator, not a severity model. For seasonal panels, counts and exposure are summed within each study season before division; the seasons partition the All year result. In the seasonal rate figures, mean-wind intervals at or above 15 m/s and gust intervals at or above 20 m/s are combined by summing counts and exposure before division. All year retains the fine bins.

Same-day allocation sensitivity. The alternative allocation uses weather observed on each actual counter-day:

$$\hat{V}_b^{\text{day}} = \sum_j \sum_d C_{jd} L_j \frac{M_{jdb}}{1020}, \quad R_b^{\text{day}} = \frac{O_b}{\hat{V}_b^{\text{day}}} 10^6,$$

where M_{jdb} is the number of observed minutes in bin b during 07:00–24:00. Traffic is assumed uniform over this window. Weather is taken from the nearest qualifying station available at each timestamp; missing weather time contributes no allocated exposure. Eligible positive-traffic days without accidents still contribute. Unlike monthly allocation, this method retains only weather-covered exposure rather than the full daily total. It therefore depends more strongly on the weather composition of individual days, completeness of observations and the assumption about traffic within the day. Sparse extreme-weather days can have greater influence. It is retained as a sensitivity, not a replacement for the main monthly-frequency denominator. The accident numerator still uses actual event-time weather in the linked sample.

The existing output supports mean wind, gust and temperature, with disjoint injury groups sharing exposure in each bin and season. Seasonal wind and gust tails are combined for display as specified in Results, while All year keeps the finer bins. This is the traffic-rate analysis's exposure-allocation sensitivity, distinct from the older allocated daily-counter regression below.

Table 3.8 summarises the two exposure constructions. Monthly allocation is the main method because only daily traffic totals are observed: typical station-month weather provides a stable allocation while retaining actual day-to-day traffic variation. It depends less on isolated storm days and missing observations within a particular day. Same-day allocation has the useful advantage of using realised weather, but distributing the full daily traffic total using that day's observed weather requires an additional assumption about traffic within the day. Neither method observes hourly exposure.

Table 3.8: Comparison of the main monthly-frequency allocation and same-day sensitivity.

	Monthly-frequency main method	Same-day sensitivity
Traffic and road input	Observed daily count $\times$ rural road length	Same
Weather allocation	Typical pooled station-month distribution	Weather observed on the actual date
Accident classification	Actual accident-time weather	Actual accident-time weather
Accident-free days	Included when eligible	Included where weather defines exposure
Hourly traffic observed?	No	No
Main advantage	Stable weather-bin allocation	Responds to realised daily weather
Key limitation	Monthly weather approximates exposure	Daily total allocated by observed weather; missing time excluded
Role	Main traffic-rate analysis	Sensitivity to allocation choice

3.3.5 Supporting Matched-Time Comparison

The supporting matched-time analysis uses a time-stratified case-crossover design in which each accident forms one matched set. The accident time is the case time, and the control times are the other occurrences of the same weekday and clock time in the same calendar month and year. For example, a Tuesday accident at 14:20 is compared with the other Tuesdays at 14:20 in that month. Mean-wind and gust controls use the accident's matched wind station; temperature controls use its independently matched temperature station. Control weather must satisfy the same five-minute observation-time rule, and both station matches are restricted to 20 km. Conditional logistic regression compares weather at the accident time with weather at its matched control times. This time-stratified referent strategy was selected to avoid overlap bias and to control calendar time without selecting controls from outside the accident month (Janes et al., 2005).

The model asks whether weather at an accident time differs from weather at comparable non-accident times. The matching holds station, year, month, weekday, and clock time fixed, but not unusual traffic, road conditions, or other factors that change on a particular day. Continuous effects are reported per 5 m/s or 5°C together with categorical estimates.

Mean wind and temperature are also entered together as categorical variables to assess whether the wind estimate changes after including temperature at the same case and control times. Daylight is calculated at the accident location for the accident date and the same matched dates. Because month, weekday, and clock time are held fixed, only strata that change between darkness, civil twilight, and daylight contribute information.

Seasonal differences in the mean-wind association are tested in a separate matched-time model. Mean wind is grouped as 0–10, 10–15, and at least 15 m/s for stability. The model includes interactions between wind interval and season, and a likelihood-ratio test compares it with a model having one common wind association across seasons. The omnibus test is used instead of judging seasonal differences from separate confidence intervals.

A separate logistic model describes serious or fatal rather than minor injury among accidents with complete wind and temperature matches. It includes broad categories of mean wind, temperature, daylight, time of day, and season. The result concerns injury severity among recorded accidents. It is not a model of whether an accident occurs and does not replace the traffic analyses.

3.3.6 Supporting Allocated Daily-Counter Model

An additional daily-counter model uses observed 24-hour totals but estimates how those totals are distributed across wind intervals within each day. Directional channels are first summed to one count for each physical counter and date. Each 2019–2024 accident then requires a counter on the exact registered road section and year, positive traffic on the accident date, and a qualifying accident-time mean-wind observation. For counter c , date d , and wind interval j , the daily count is allocated as

$$\hat{Q}_{cdj} = Q_{cd} \frac{N_{cdj}}{N_{cd}},$$

where N_{cdj} is the number of valid 10-minute observations in interval j and N_{cd} is the total number of valid observations that day. Accidents are classified by accident-time mean wind from the same station used for the counter-day denominator. Estimated traffic and accidents are aggregated by counter, year, and wind interval, and a conditional Poisson model gives each counter–year its own intercept. The allocation assumes uniform traffic across valid observation times and is therefore an estimated within-day split rather than observed hourly exposure. Selection and sensitivity checks for this 762-accident supporting model are reported in the appendix.

4 Results

Results are presented in the order of the three research questions. First, the primary weather-frequency analysis compares observed accidents with counts expected from local weather frequency. Second, an approximate traffic correction asks how that comparison changes when the traffic response measured at daily counters is applied to the primary denominator. Third, the monthly-frequency counter analysis estimates accident rates per vehicle-kilometre in the smaller 2019–2024 traffic-linked sample. Matched-time and additional traffic models are reported afterwards as supporting checks. Because these analyses use different samples and denominators, their numerical estimates are not interpreted as repeated estimates of the same quantity.

4.1 Study Sample and Coverage

Wind and temperature have identical match coverage in the rural injury accident sample: 6,259 of 6,414 accidents (97.6%) for each measure. The annual match percentages remain high throughout the study (Figure 4.2).

Figure 4.1 shows a decline spanning the financial-crisis period and a lower count in 2020 during the COVID-19 period, followed by a partial recovery. The dip in 2014 is also visible. The official accident report documents the reporting system and earlier changes but does not establish a 2014 reporting change that explains this dip (ITA, 2015). These time patterns alone do not identify their causes or separate changes in travel from changes in accident risk.

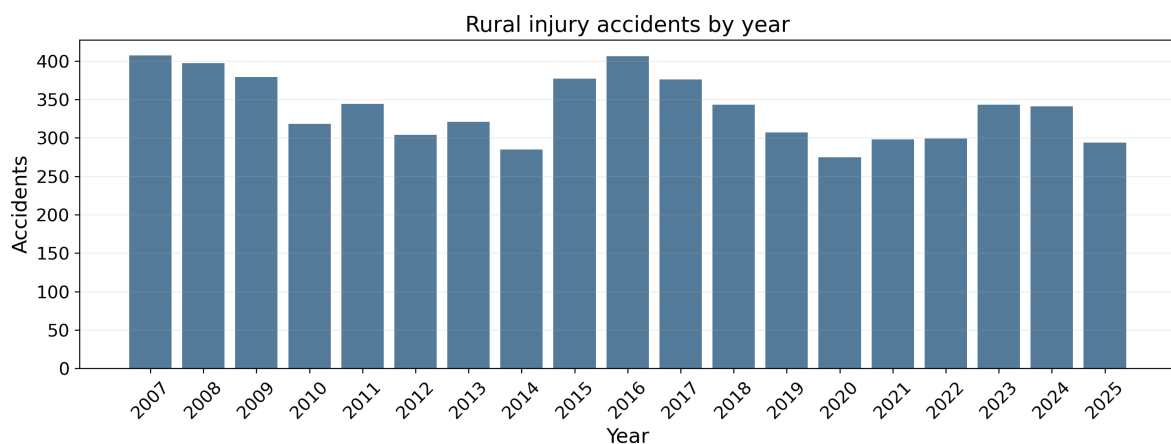


Figure 4.1: Annual rural injury accident counts, 2007–2025.

The traffic-based analysis uses a smaller, counter-linked population. Table 4.1 follows selection to the final 694 accidents. All 83 exclusions at the final step lack a counter-day record on the accident date; none is removed because its recorded daily count is zero.

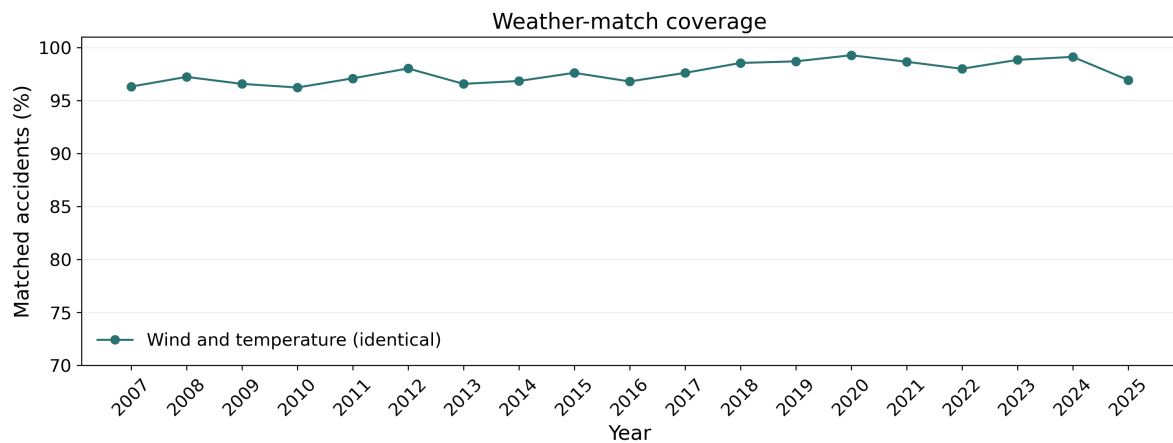


Figure 4.2: Annual weather-match coverage among rural injury accidents, 2007–2025. Matches require a qualifying observation within 20 km and five minutes; wind and temperature percentages coincide in each year.

Table 4.1: Selection for the traffic-based rate analysis. The first row includes urban and rural injury accidents; the rural restriction is applied in step (b). The final set contains minor, serious and fatal injury accidents and is distinct from the supporting allocated daily-counter sample.

Step	Removed	Remaining
All injury accidents, 2007–2025	–	17,045
a) Restrict to 2019–2024	11,849	5,196
b) Exclude urban accidents	3,333	1,863
c) Require counter-section for road/year	994	869
d) Require successful road-location assignment	21	848
e) Exclude 00:00–07:00	67	781
f) Require usable wind and gust (≤ 20 km, ± 5 min)	4	777
g) Require daily traffic count	83	694

Notes: The 21 road-location failures in step (d) comprise five records without usable road geometry, 15 located more than 100 m from their registered road, and one projected outside the counter sections. For step (f), the 20 km weather distance is measured from the counter-section location rather than the accident coordinate; valid mean wind and gust are required within five minutes of the recorded accident time.

4.2 Weather-Frequency Baseline

Figures 4.3–4.5 show the primary weather-frequency comparisons for the full year and the four study seasons. Blue bars show minor-injury accidents and red bars show serious or fatal injury accidents; the groups do not overlap. The grey horizontal line marks $O/E = 1$, where observed accidents equal the number expected from local weather frequency. Values above one indicate over-representation relative to that frequency, not a measured increase in risk per vehicle-kilometre.

4.2.1 Mean Wind

Figure 4.3 shows the main result: O/E is close to one at low and moderate mean wind speeds and rises in the upper intervals. At 20 m/s or above, the all-injury O/E is 2.15. Both injury groups show this upper-wind pattern, but the serious or fatal group contains few accidents. Differences between the bars should therefore not be read as evidence that wind changes injury severity.

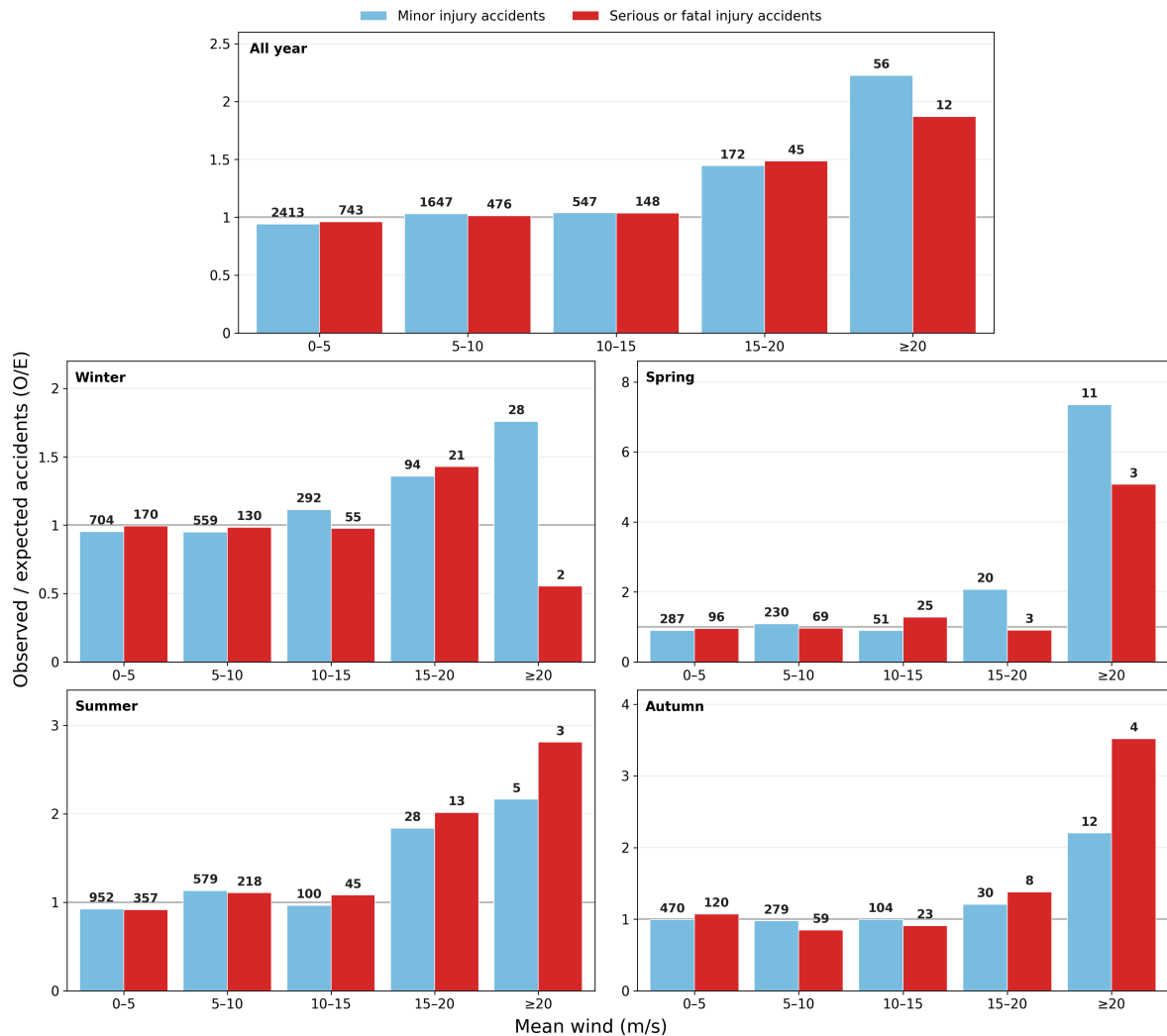


Figure 4.3: All-year and seasonal mean-wind O/E, 2007–2025. Blue bars show minor injury and red bars show serious or fatal injury. Expected counts are calculated from local station–season weather frequency separately for each group; labels above bars are observed accident counts. The grey reference line is $O/E = 1$. Seasonal upper-wind bars should be interpreted with their small observed counts.

The seasonal panels are less stable than the pooled result. In particular, some of the largest seasonal O/E values are based on only a few accidents, so they do not support ranking seasons or injury groups. The 5 m/s intervals are used as descriptive bins rather than physical wind thresholds.

4.2.2 Wind Gust

Wind gust shows a similar upper-tail pattern in Figure 4.4, with all-injury O/E 3.34 at gusts of at least 30 m/s. Both injury groups are above one in this interval. Gust and mean wind are related measurements from the same observation, so their agreement supports a general high-wind pattern rather than an independent replication. Seasonal extremes remain sparse.

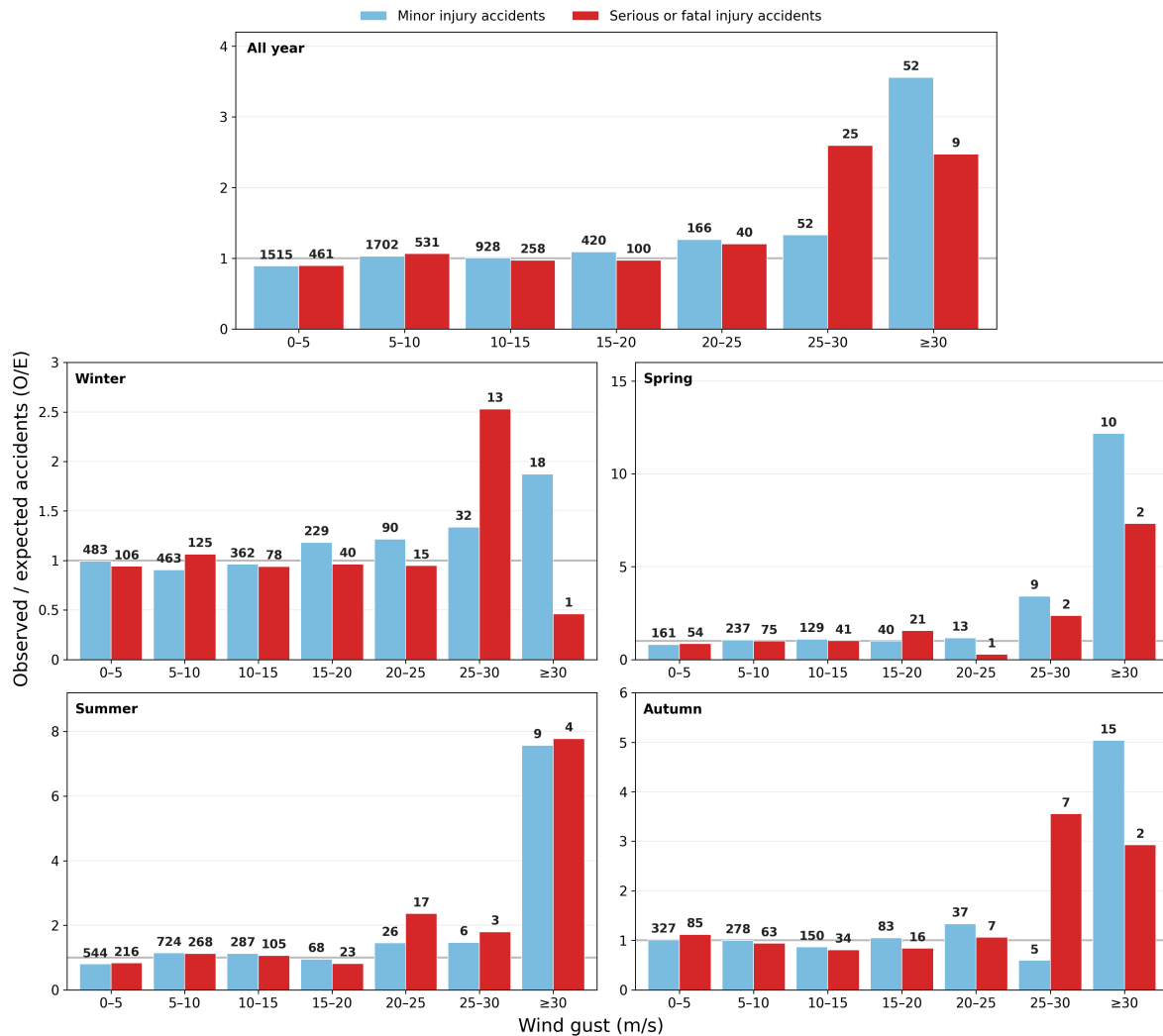


Figure 4.4: All-year and seasonal wind-gust O/E. Gust is taken from the same 10-minute observation as the accident-time mean wind and is not a daily maximum. Blue bars show minor injury and red bars show serious or fatal injury; labels above bars are observed accident counts. The grey reference line is O/E = 1. Several seasonal upper-gust cells contain few accidents and are therefore descriptive.

4.2.3 Temperature

Temperature has a non-monotonic pattern in Figure 4.5: O/E is lower at moderate temperatures and higher in some colder and warmer intervals. Both injury groups show this broad shape, but seasonal extremes contain few accidents. The finding is secondary and does not establish a causal temperature effect.

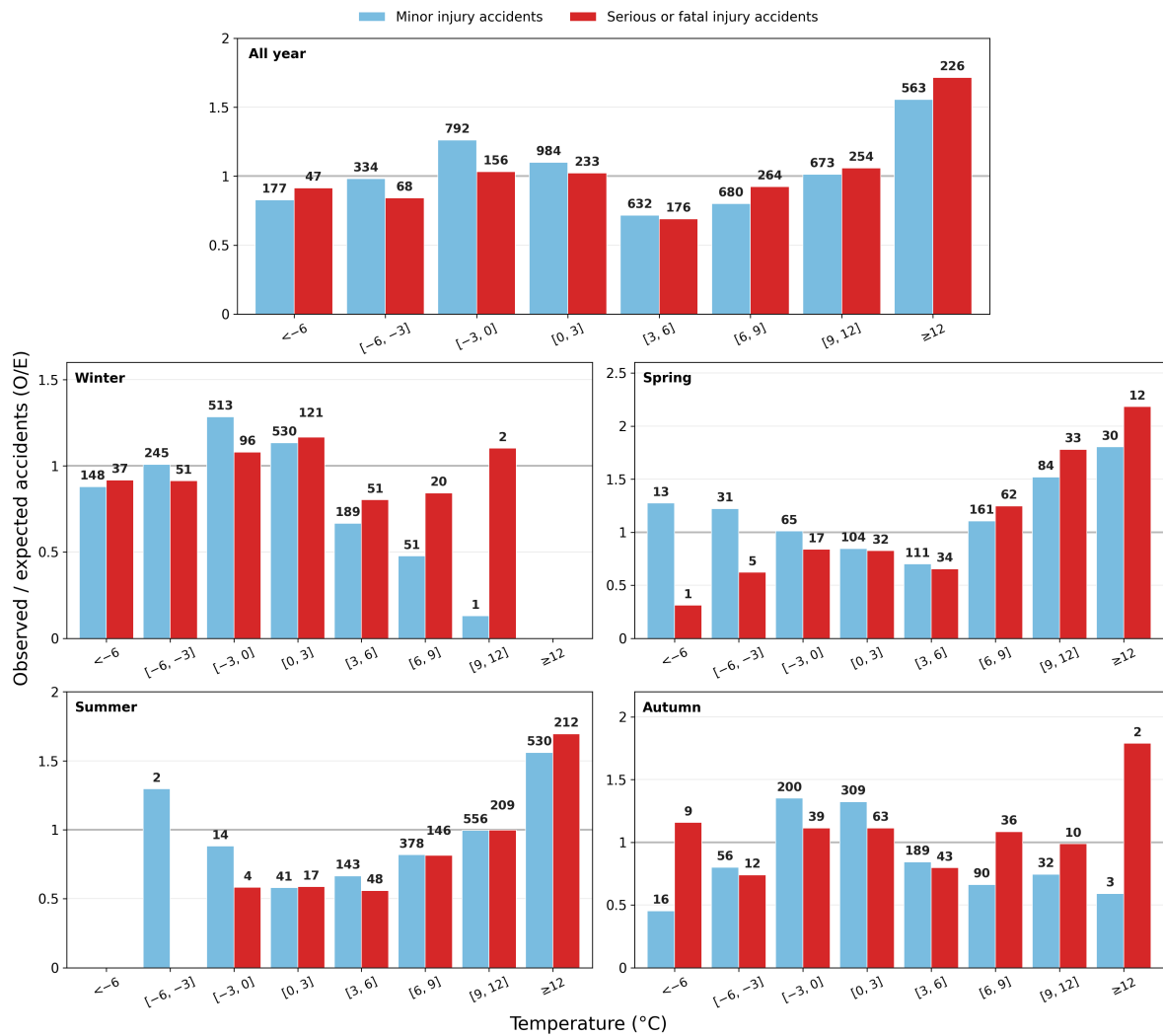


Figure 4.5: All-year and seasonal temperature O/E, standardised within matched temperature station and study season. Blue bars show minor injury and red bars show serious or fatal injury; labels above bars are observed accident counts. The grey reference line is $O/E = 1$. Temperature intervals follow the display convention defined in Methods.

Adding year to the station–season standardisation changes the selected O/E estimates only slightly. The pre-specified comparison is therefore retained as the weather-only analysis.

4.3 Traffic Response and Approximate O/E Correction

4.3.1 Traffic Behaviour and the Baseline Limitation

Figure 4.6 shows the traffic response underlying the correction. Values below 100% indicate less traffic than the counter’s calendar expectation, not fewer accidents. The upper wind and gust intervals show reductions. The temperature response is much smaller across the displayed intervals.

This motivates an approximate correction under the assumption that the traffic response observed at the selected counters is informative for the wider road network. It does not establish the magnitude of a national per-vehicle-kilometre contrast, because the traffic response is estimated over a shorter period and from a geographically selective set of counters.

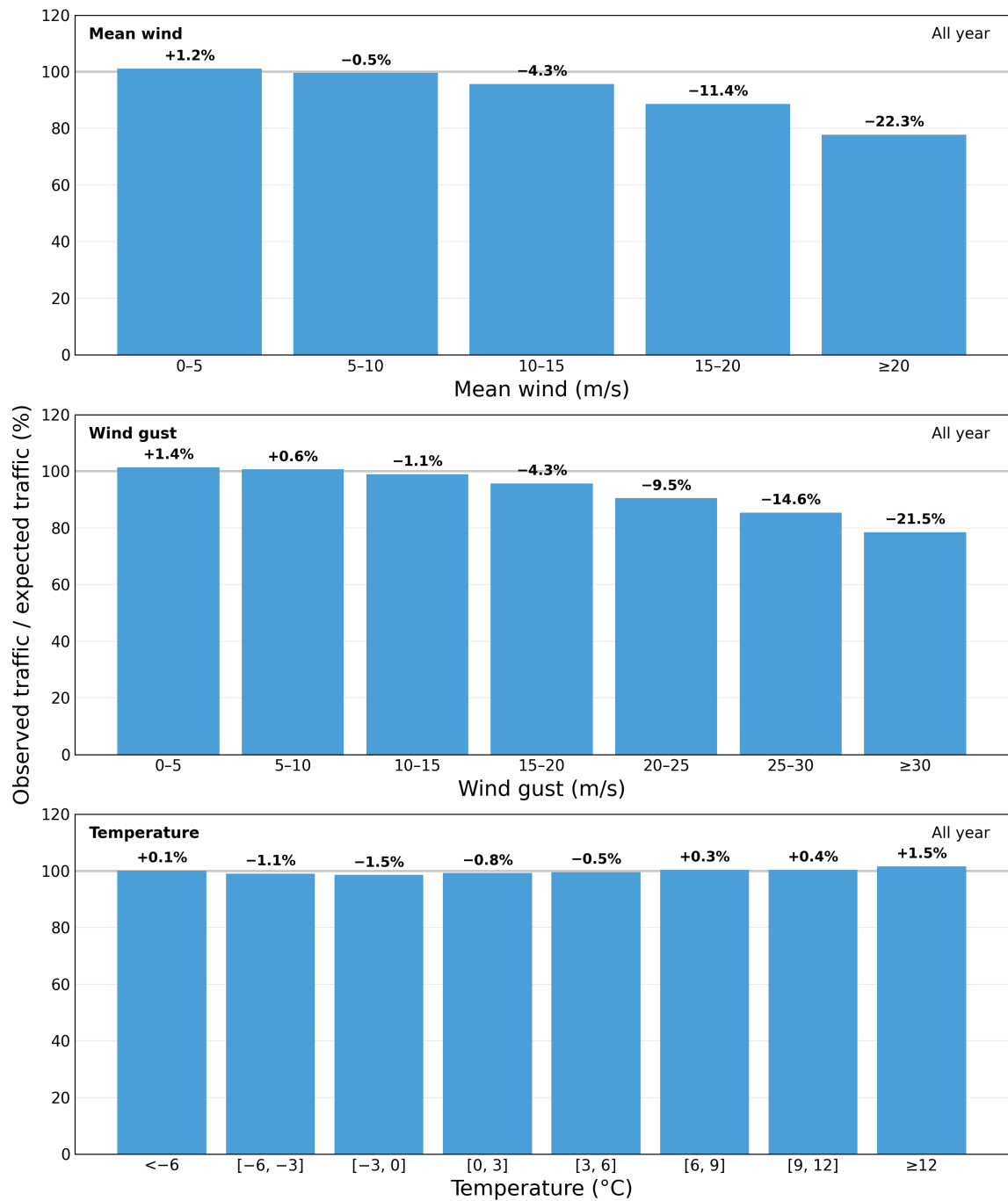


Figure 4.6: Observed daily traffic relative to expected traffic at counter sections, 2019–2024, pooled across seasons. The expectation uses the same section, year, month and weekday among positive-count days. Counts are allocated by observed 07:00–24:00 weather-bin minutes; bars show the ratio of pooled allocated observed to expected traffic. The line marks 100%; labels show percentage changes, not accident counts.

4.3.2 Full-Period Correction

The primary O/E uses weather time rather than vehicle exposure. Daily-counter data from 2019–2024 show that traffic is lower in the strongest wind and gust intervals, so the approximate correction applies the estimated counter response to the primary station–season denominator as a sensitivity analysis. The observed accident counts remain unchanged; only the expected distribution is reweighted.

The change is visible directly in Figures 4.7 and 4.8: upper-bin O/E rises from 2.15 to 2.71 for mean wind and from 3.34 to 4.20 for gust. Outlined bars show the weather-only analysis and filled teal bars show the approximate correction for the same all-injury population. Thus, accounting for lower traffic strengthens the high-wind contrast. The validated correction is available only for All year.

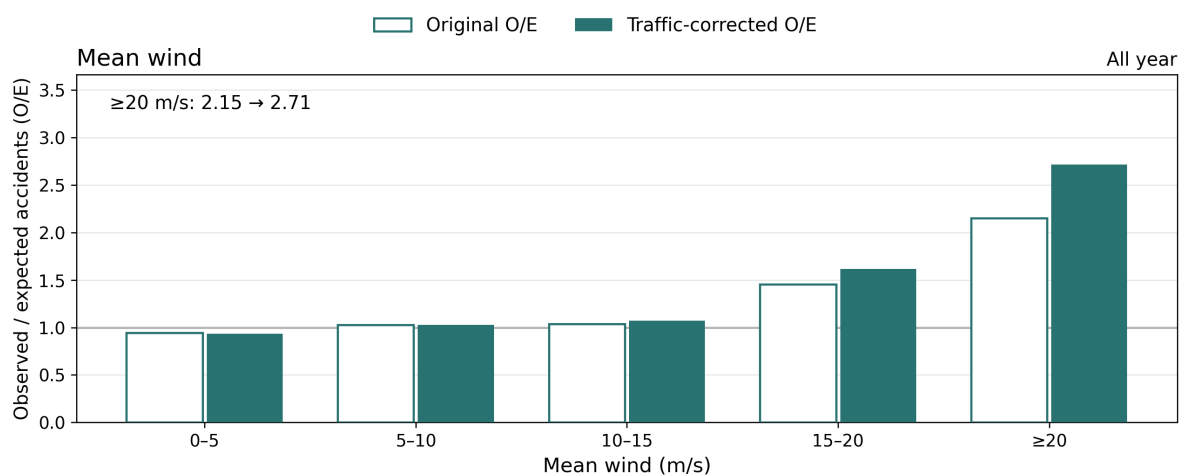


Figure 4.7: Original and approximately traffic-corrected mean-wind O/E for all qualifying rural injury accidents, 2007–2025, All year. Outlined bars use weather frequency; teal bars reweight it using the 2019–2024 counter response. The annotation highlights the upper-bin change; the reference line is $O/E = 1$. Extrapolation beyond the counters and absence of hourly traffic make this a model-based sensitivity analysis.

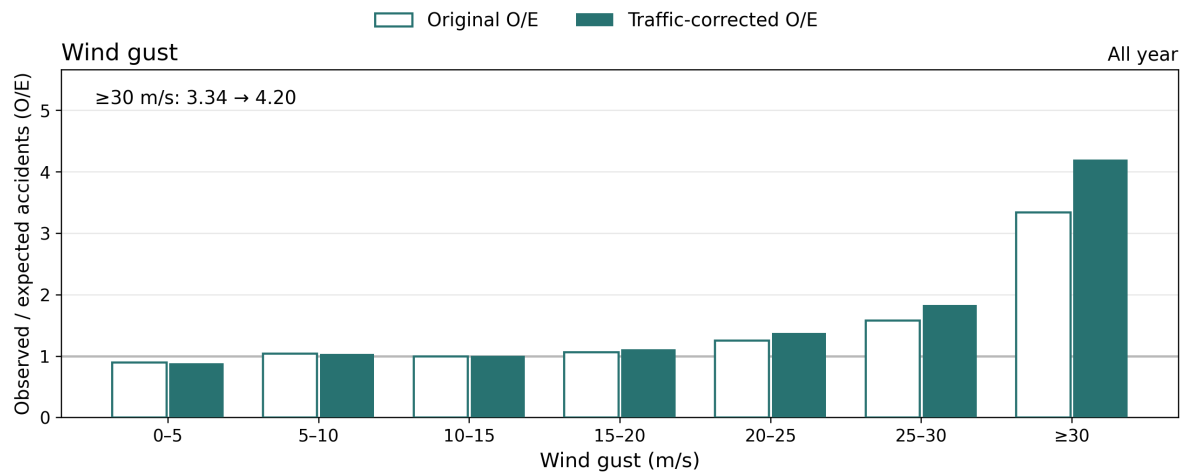


Figure 4.8: Original and approximately traffic-corrected wind-gust O/E for all qualifying rural injury accidents, 2007–2025, All year. Outlined and teal bars use the original and reweighted expected counts, respectively. The annotation highlights the upper-bin change; the reference line is $O/E = 1$. The 2019–2024 daily-counter response is extrapolated to the wider population and does not identify hourly traffic exposure.

Figure 4.9 retains the corrected comparison separately for minor injury and serious or fatal injury, using the same two groups as the weather-only comparison. This shows whether the corrected upper-wind pattern is shared across injury categories. Differences between the bars are not a test of severity, particularly where the serious or fatal counts are small.

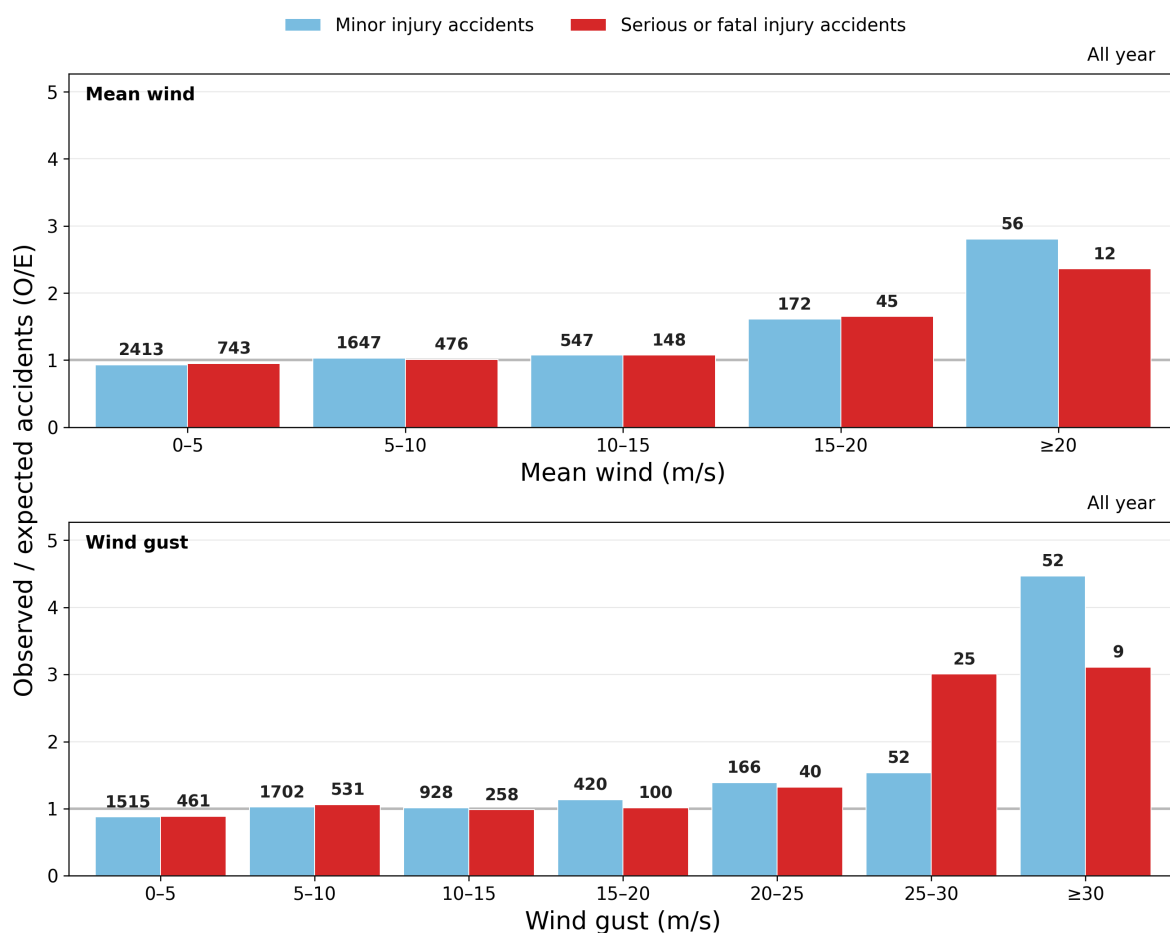


Figure 4.9: Approximately traffic-corrected mean-wind and gust O/E, 2007–2025, All year. Blue and red bars represent minor injury and serious or fatal injury. Each group’s expected count is reweighted using the 2019–2024 counter response; labels give observed counts. The line marks $O/E = 1$. These disjoint groups complement the all-injury original–corrected comparison; the same extrapolation and hourly-traffic limitations apply.

4.4 Counter-Based Accident Rates

The third stage estimates accident rates in the smaller sample linked to daily traffic counters. It includes 694 accidents, 533,649 eligible counter-section days and approximately 5.63 billion estimated vehicle-kilometres. Table 4.2 gives the accident counts, exposure and rates for each wind interval. The main method uses monthly weather frequencies; the same-day allocation sensitivity in the following section tests that choice. Both constructions are defined in the Methods.

Table 4.2: Monthly-frequency daily-counter rates, 2019–2024. Estimated exposure equals the observed daily traffic count times rural section length, allocated across weather intervals using the assigned station’s pooled 2007–2025 calendar-month weather frequency for 07:00–24:00.

Measure	Interval (m/s)	Accidents	Million vehicle-km	Accidents per million vehicle-km
Mean wind	0–5	344	3011.5	0.114
Mean wind	5–10	210	1963.5	0.107
Mean wind	10–15	96	539.9	0.178
Mean wind	15–20	33	98.9	0.334
Mean wind	≥20	11	16.5	0.667
Gust	0–5	205	1840.9	0.111
Gust	5–10	238	2133.5	0.112
Gust	10–15	129	1085.1	0.119
Gust	15–20	70	400.3	0.175
Gust	20–25	29	123.3	0.235
Gust	25–30	12	34.5	0.348
Gust	≥30	11	12.7	0.866

Rates rise markedly in the upper wind intervals. At mean wind of at least 20 m/s, the rate is 5.83 times the 0–5 m/s rate; for gusts of at least 30 m/s, the contrast is 7.78. Only 11 accidents contribute to each upper interval, so these large differences are imprecise. The selective counter coverage also limits their generalisability.

4.4.1 All Year Rates

Figure 4.10 shows the pooled wind and gust rates. The light-blue and red components share estimated exposure, so their stack height is the all-injury rate. Counts identify how many accidents contribute; the bars are not a formal comparison of injury severity.

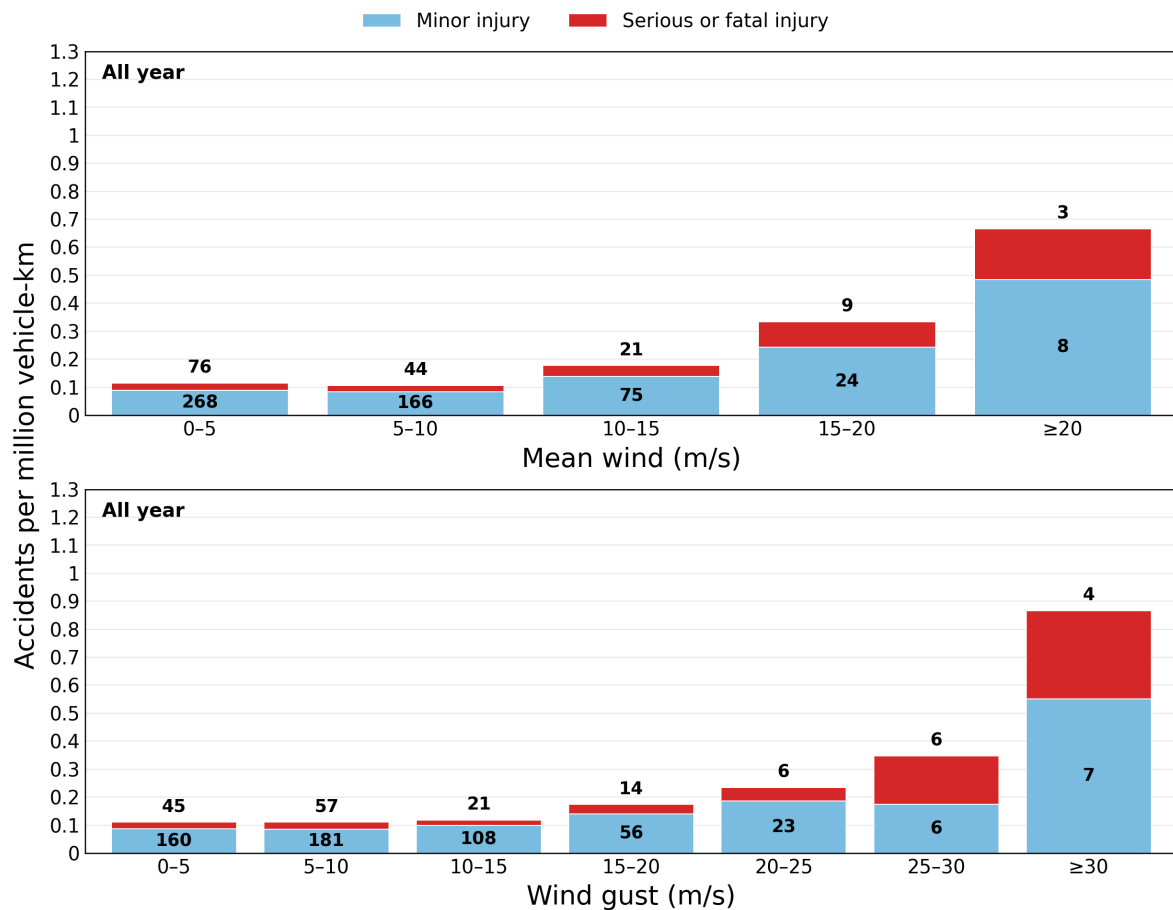


Figure 4.10: Rural injury-accident rates per million vehicle-km, 2019–2024, All year. Event-time wind and gust classify the accidents; daily traffic, rural length and pooled station-month weather frequencies define estimated exposure. Light blue is minor injury; red is serious or fatal injury. The stacked components sum to the all-injury rate; positive labels give counts. Full daily traffic is allocated using 07:00–24:00 weather frequencies; hourly traffic is unobserved.

4.4.2 Mean Wind by Season

Figure 4.11 shows how the mean-wind pattern varies by season. Seasonal intervals combine the upper tail at 15 m/s; counts and exposure are summed before calculating each rate. Blue and red stacks show the minor and serious or fatal components of each all-injury rate, sharing the same exposure. Positive counts identify sparse cells; segment sizes are not a formal severity comparison. These panels are descriptive: a season with no observed accident in an interval is not necessarily free of risk.

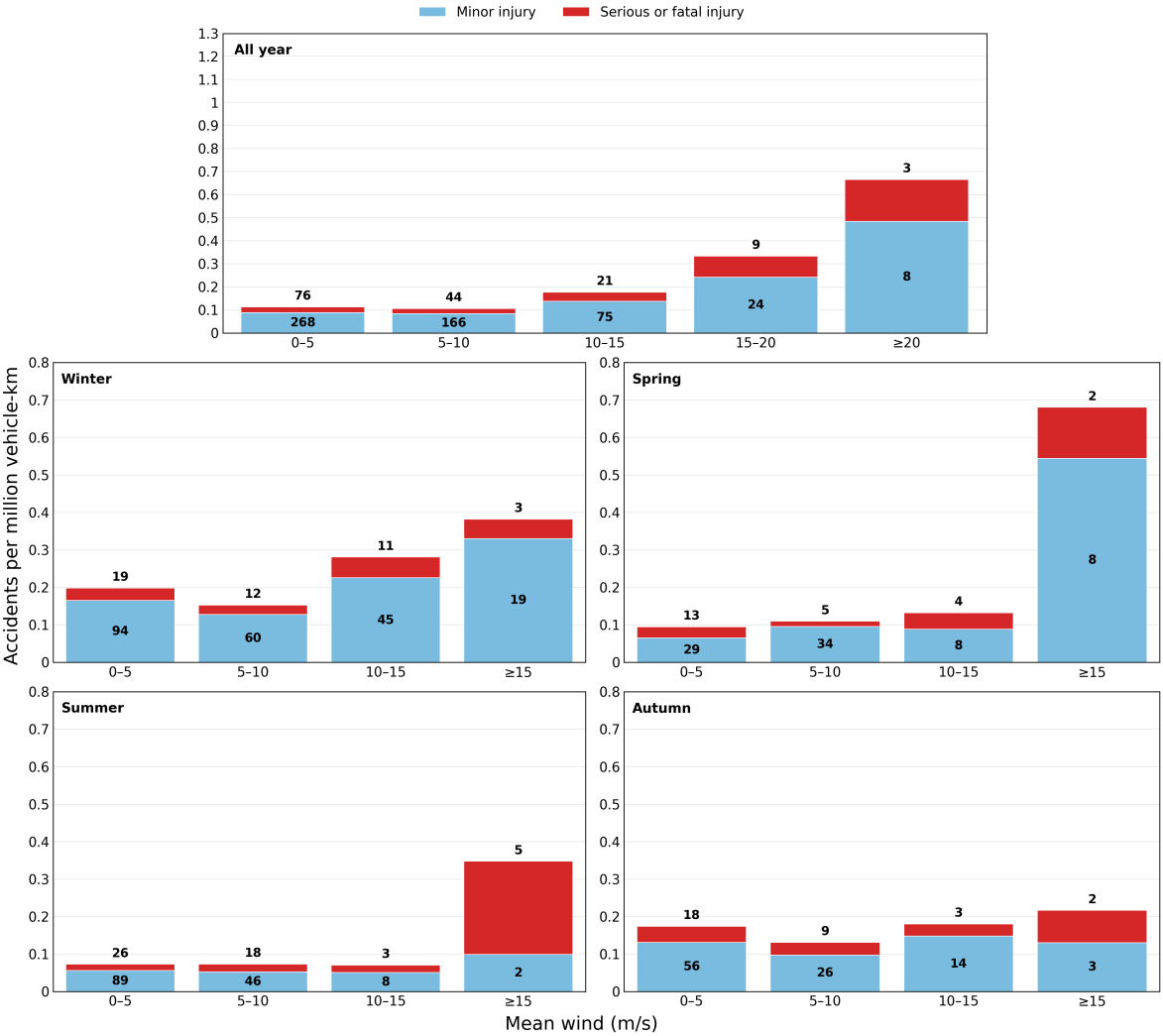


Figure 4.11: Rural injury accident rates by mean wind, 2019–2024. Light blue and red show disjoint minor and serious or fatal injury rates sharing estimated vehicle-km from daily traffic, rural length and pooled monthly weather. All year retains fine bins; seasonal tails combine intervals from 15 m/s. Positive labels give accident counts. Sparse cells and unobserved hourly traffic limit interpretation.

4.4.3 Wind Gust by Season

Figure 4.12 shows the corresponding stacked gust pattern, with seasonal gust intervals combined at 20 m/s. The upper-bin accidents are spread thinly across seasons, so the panels do not establish seasonal differences. The main monthly-allocation method supports mean wind and gust only. Temperature is available in the same-day sensitivity below.

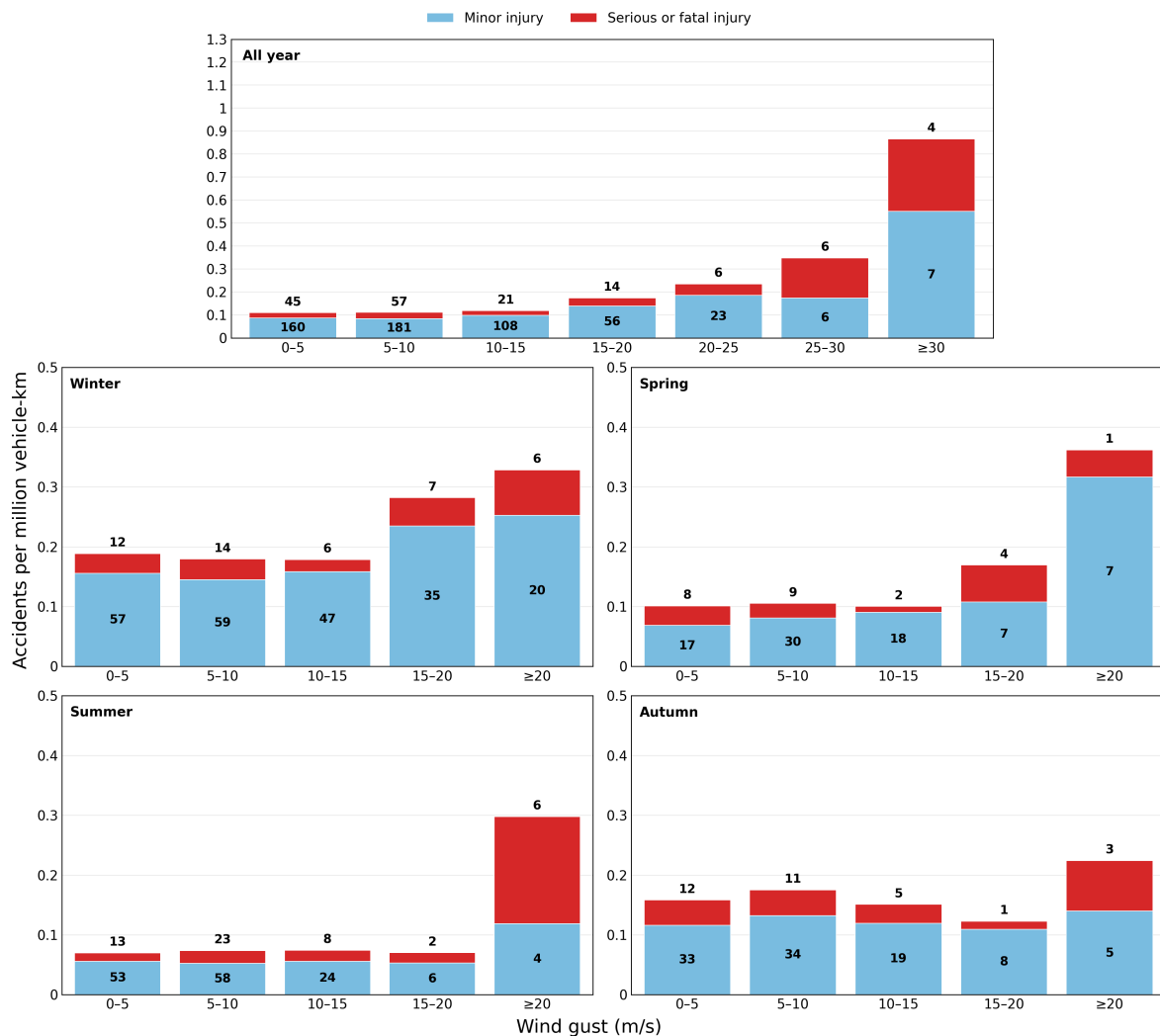


Figure 4.12: Rural injury accident rates by gust, 2019–2024. Light blue and red show disjoint minor and serious or fatal injury rates sharing estimated vehicle-km from daily traffic, rural length and pooled monthly weather. All year retains fine bins; seasonal tails combine intervals from 20 m/s. Positive labels give accident counts. Sparse cells and unobserved hourly traffic limit interpretation.

A stratified diagnostic gives smaller upper-bin contrasts when the comparison is made within counter-section-year-season strata: O/E is 3.39 for mean wind and 4.84 for gust. This suggests that sample and road-exposure composition contribute to the larger pooled traffic-based rate ratios. The interpretation of this difference is considered in Chapter 5 rather than treated as a separate main result here.

Table 4.3 summarises the headline wind and gust results for the three main analyses. Read each estimate against its stated denominator: the table is an interpretation guide, not a ranking of estimates of one quantity.

Table 4.3: Headline mean-wind and gust results. The weather-only and corrected analyses report O/E; the traffic-based analysis reports the upper-bin rate divided by the 0–5 m/s rate. The estimates use different denominators and are not repeated estimates of one parameter. The traffic-based analysis uses a smaller counter-linked sample, with 11 accidents in each upper interval.

Parameter	Method	Comparison	Estimate	Denominator
Mean wind	Weather-only O/E	≥ 20 m/s	O/E 2.15	Local weather frequency
Mean wind	Corrected O/E	≥ 20 m/s	O/E 2.71	Traffic-reweighted weather frequency
Mean wind	Traffic-based rate	≥ 20 m/s vs 0–5	$5.83\times$	Estimated distance travelled
Wind gust	Weather-only O/E	≥ 30 m/s	O/E 3.34	Local weather frequency
Wind gust	Corrected O/E	≥ 30 m/s	O/E 4.20	Traffic-reweighted weather frequency
Wind gust	Traffic-based rate	≥ 30 m/s vs 0–5	$7.78\times$	Estimated distance travelled

4.5 Same-Day Weather Allocation Sensitivity

Figure 4.13 and the seasonal Figures 4.14, 4.15 and 4.16 present the same-day weather allocation sensitivity. These figures test sensitivity to the allocation assumption; the monthly-frequency method remains the main traffic-rate analysis. The same 694 linked accidents are classified by accident-time weather, but exposure follows the weather observed on each counter-day and excludes unobserved weather time. The resulting rates therefore have a different denominator from the monthly method. Their agreement in the broad wind pattern is more informative than a direct comparison of magnitudes.

The wind and gust seasonal plots combine upper intervals to reduce sparse cells: mean wind ends at 15 m/s or above and gust at 20 m/s or above. All year retains the finer intervals. Temperature retains its existing intervals throughout. This difference in binning must be considered before comparing panel heights; no seasonal rank or causal effect follows from the largest bar.

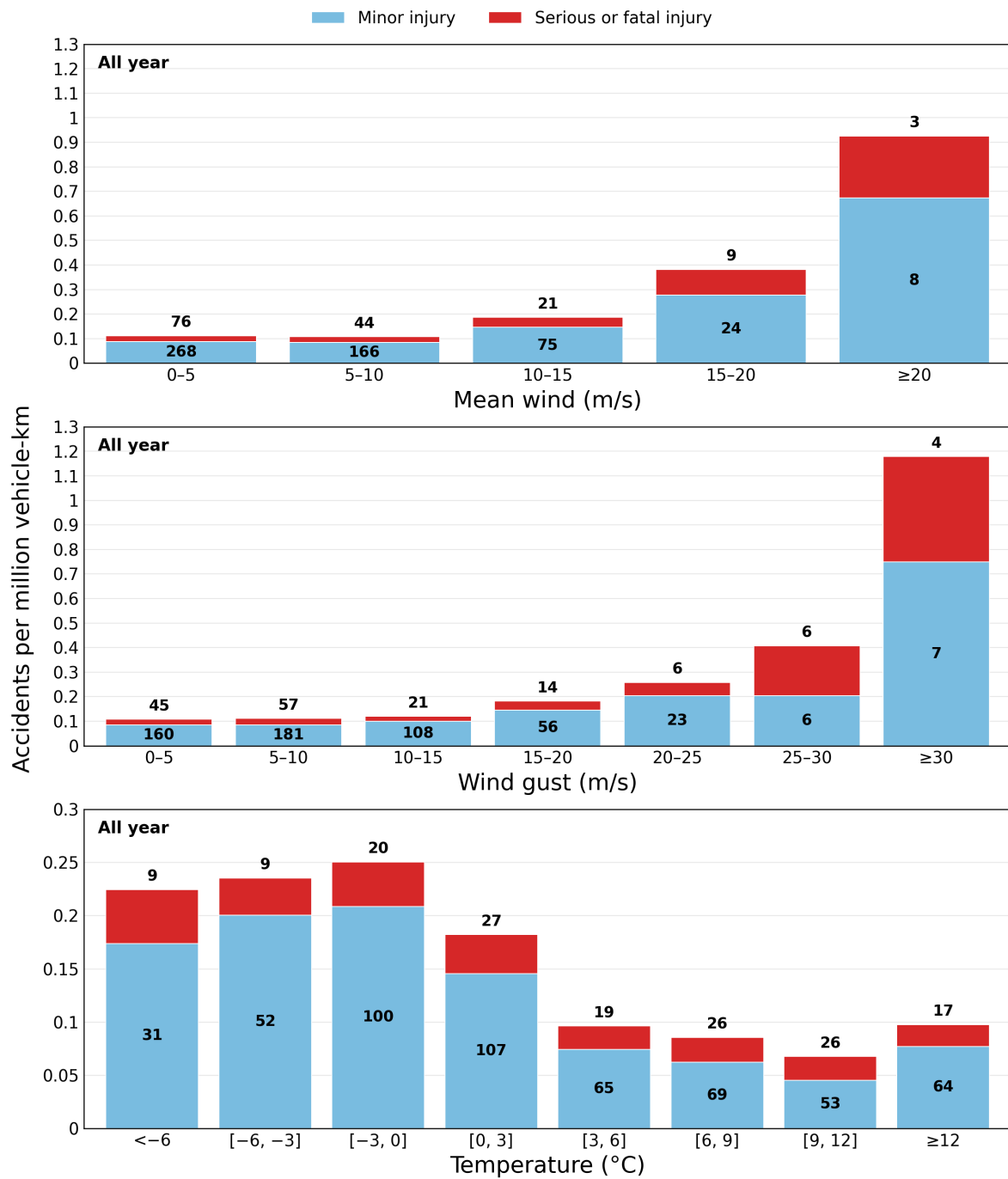


Figure 4.13: Same-day weather allocation sensitivity for rural injury accidents, 2019–2024, All year. The numerator uses event-time weather; exposure is daily traffic times rural length times observed bin minutes within 07:00–24:00 divided by 1,020. Unobserved weather minutes contribute no exposure. Light-blue minor and red serious or fatal rates share this denominator; positive labels give counts. Parameter-specific scales follow the original presentation. Daily counts do not measure hourly traffic.

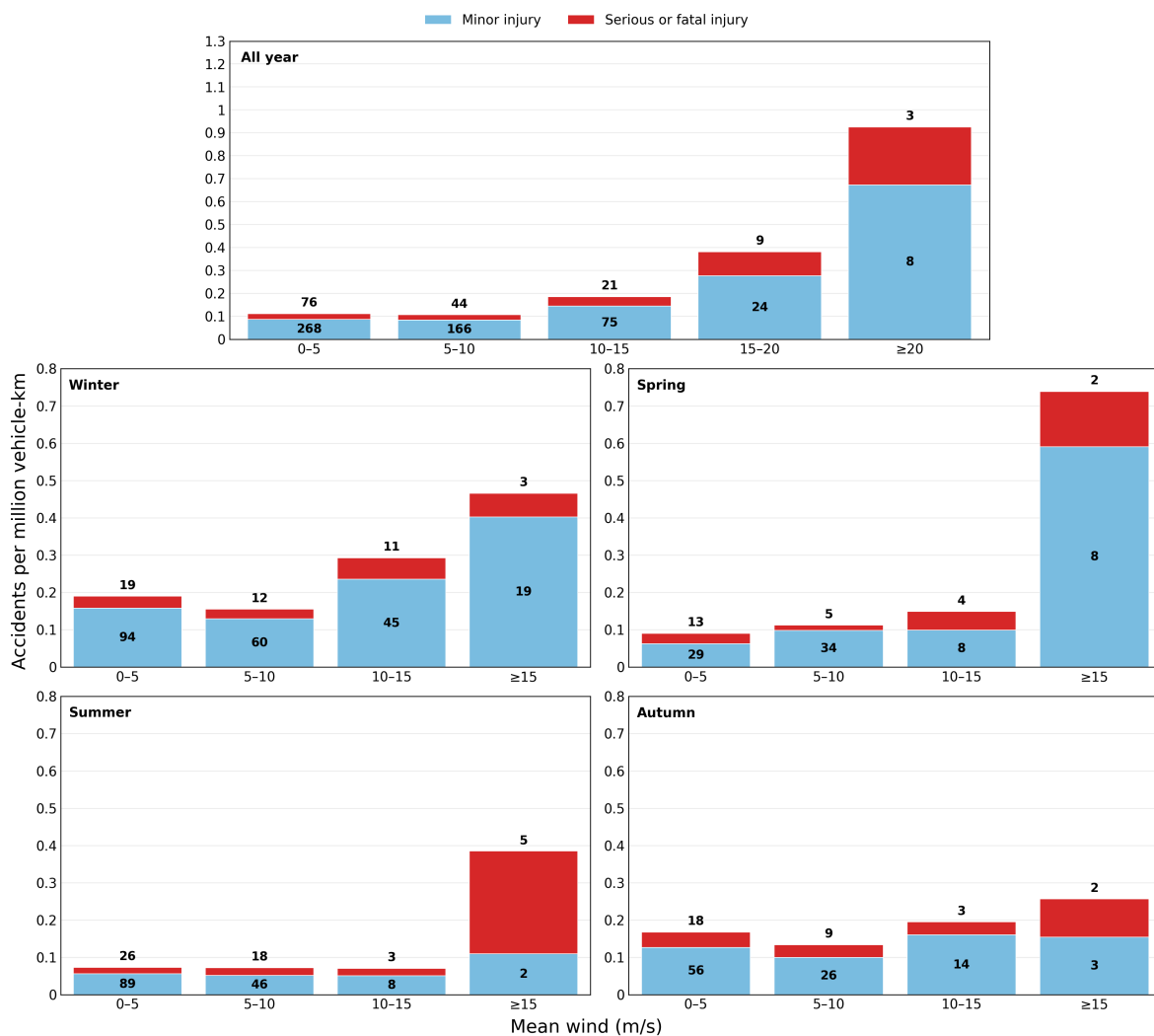


Figure 4.14: Same-day mean-wind accident rates per million vehicle-km, 2019–2024. Estimated exposure allocates daily traffic using that day's 07:00–24:00 weather, excluding unobserved weather time. Blue and red stacks show minor injury and serious or fatal injury; readable positive counts are labelled. Seasonal upper bins are combined as described in the text. This is a sensitivity to same-day rather than monthly exposure allocation.

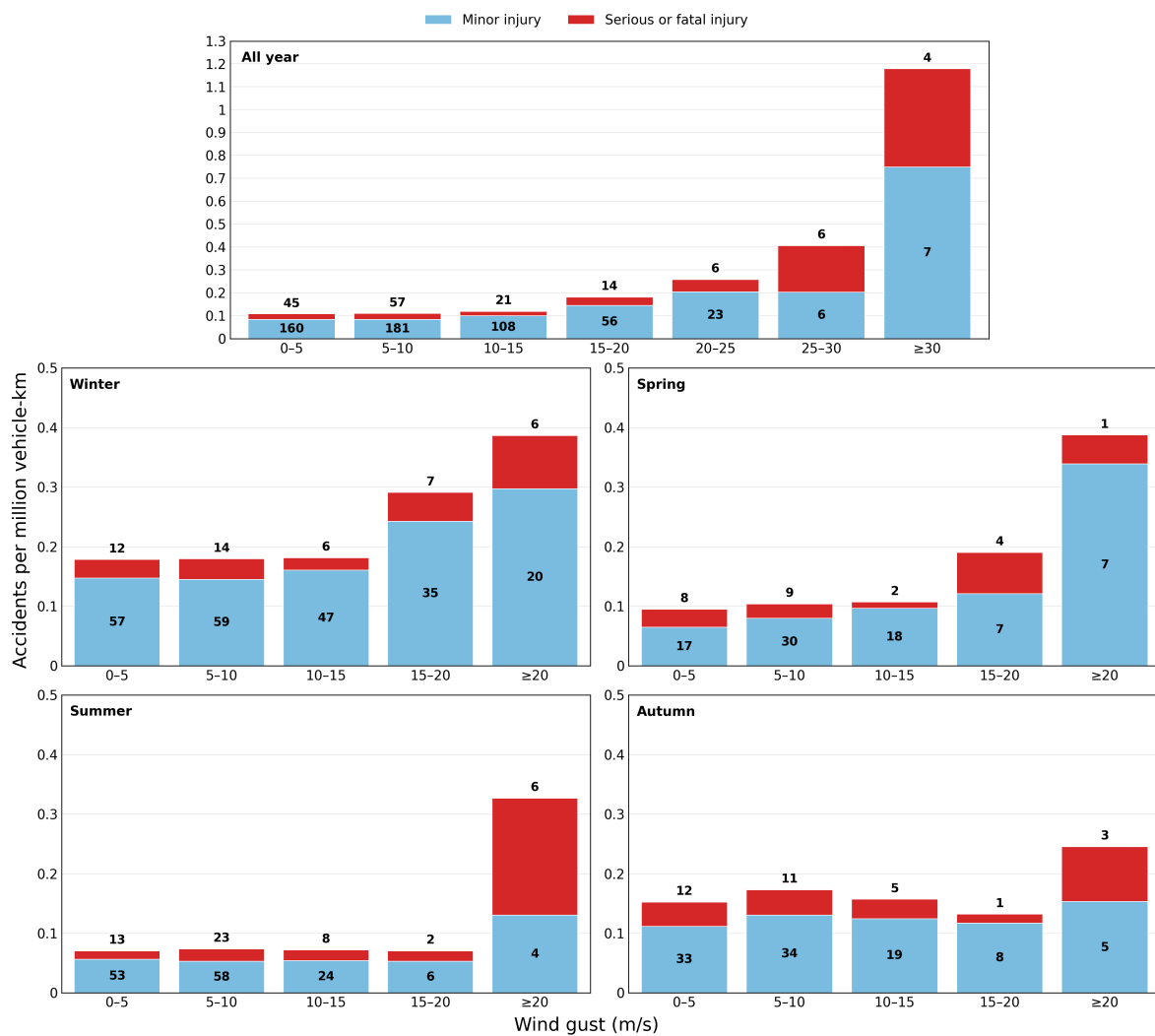


Figure 4.15: Same-day wind-gust accident rates per million vehicle-km, 2019–2024. Estimated exposure allocates daily traffic using that day's 07:00–24:00 weather, excluding unobserved weather time. Blue and red stacks show minor injury and serious or fatal injury; readable positive counts are labelled. Seasonal upper bins are combined as described in the text. This is a sensitivity to same-day rather than monthly exposure allocation.

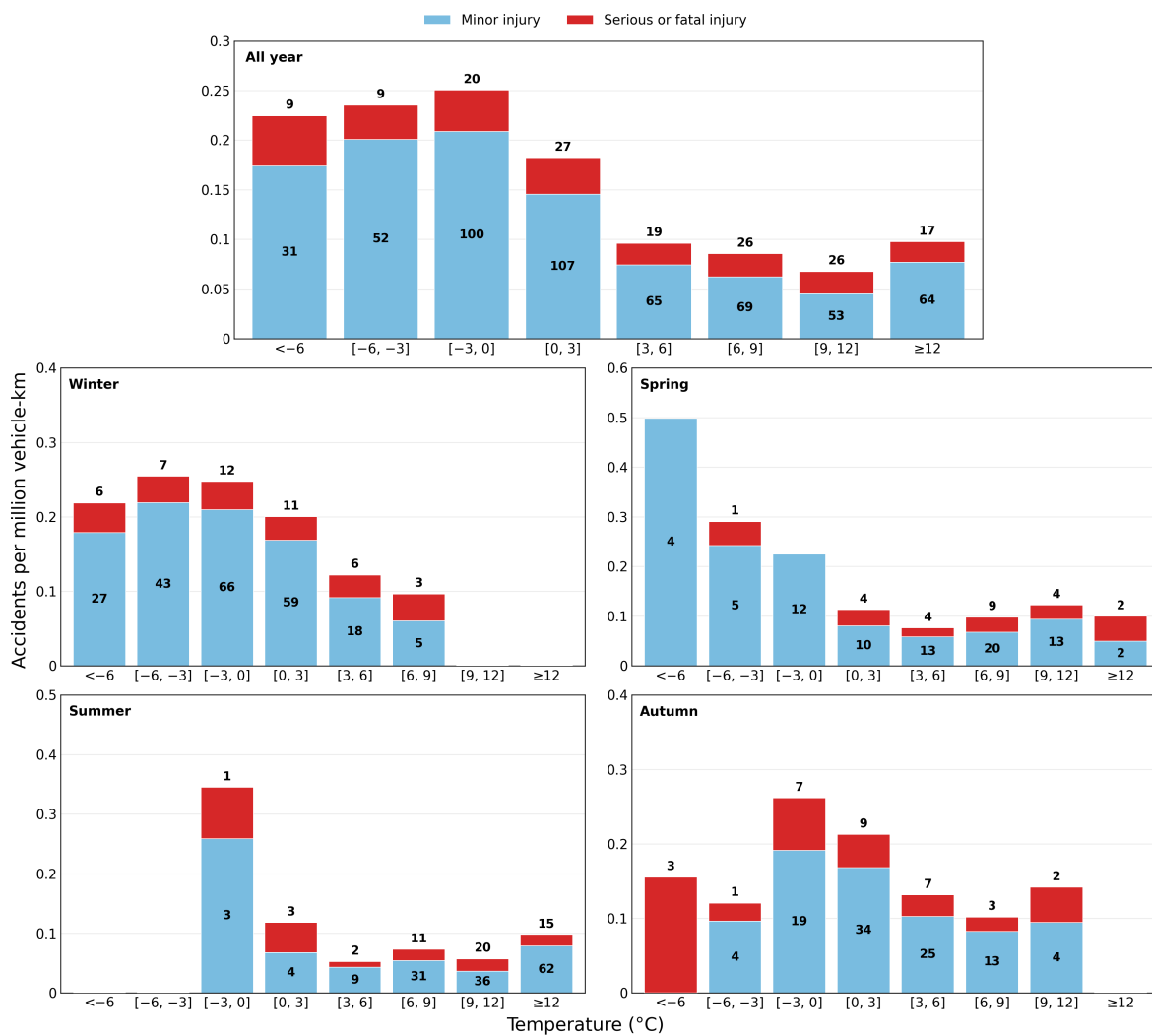


Figure 4.16: Same-day temperature accident rates per million vehicle-km, 2019–2024. Estimated exposure allocates daily traffic using that day’s 07:00–24:00 weather, excluding unobserved weather time. Blue and red stacks show minor injury and serious or fatal injury; readable positive counts are labelled. This is a sensitivity to same-day rather than monthly exposure allocation.

4.6 Secondary Robustness Checks

These checks use different statistical conditioning or distinct traffic models. The allocation sensitivity above and period comparisons below examine exposure assumptions and period choice; the following checks test other aspects of the wind association.

4.6.1 Period Comparison and Counter-Era Correction

Figures 4.17 and 4.18 show the period comparison for mean wind, gust and temperature. Each figure pools accidents and background weather over its stated years. These are All year comparisons for different periods, not seasonal estimates or replacements for the full-period weather-only figures. Upper-bin counts should be read alongside the bar heights when comparing the periods.

Figure 4.19 shows the existing correction restricted to 2019–2024, including its temperature result. Its minor-injury and serious or fatal bars can be compared with Figure 4.18, which uses the same population without traffic reweighting. This sensitivity output uses the shorter period's weather background; it must not be substituted for the approximate correction's full-period estimates. Its temperature panel concerns corrected weather-frequency O/E, not the traffic-based rate method.

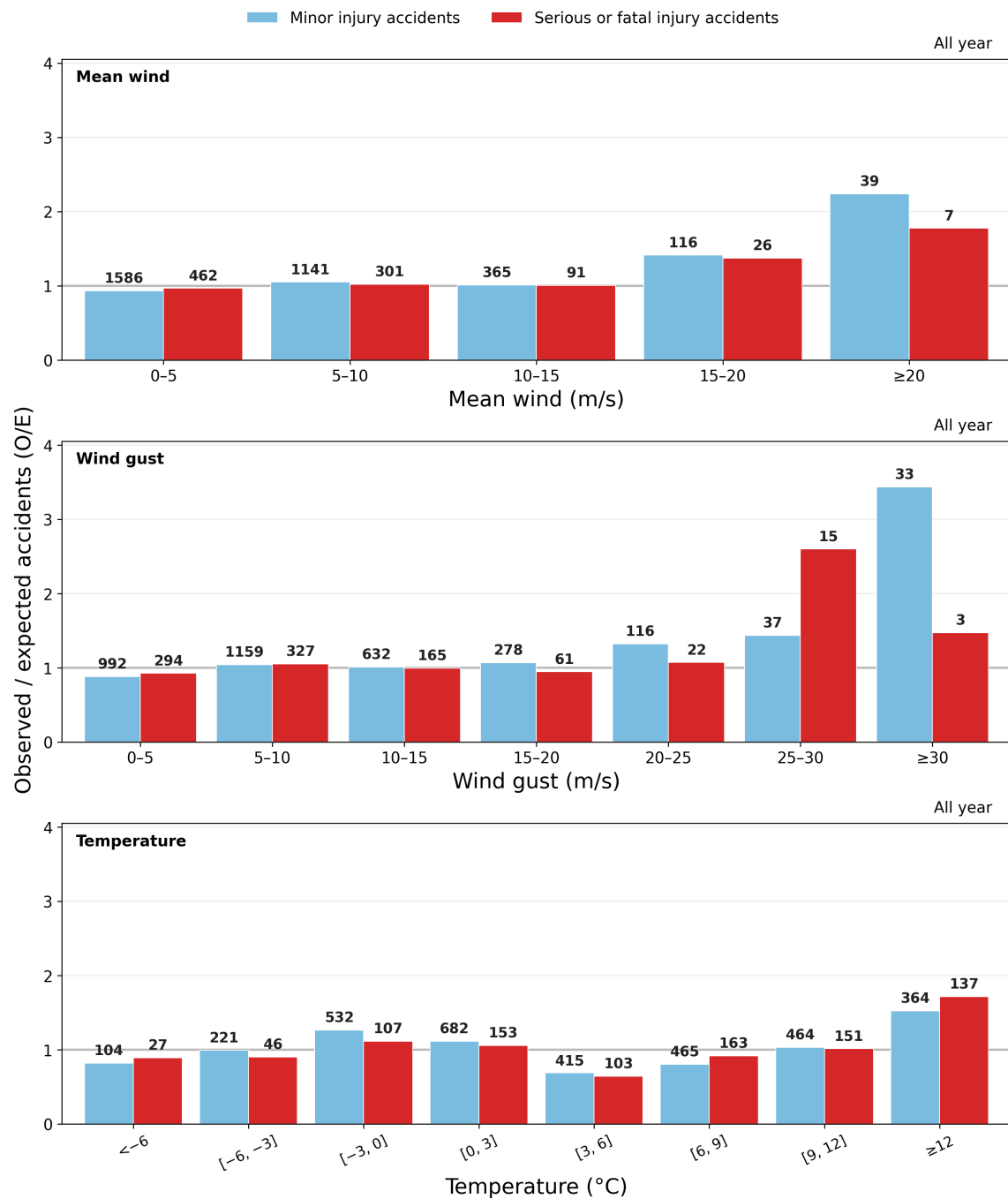


Figure 4.17: Period comparison: All year weather-frequency O/E for rural injury accidents in 2007–2018. Accident counts and station–season background weather are restricted to the same years. Blue and red bars show minor injury and serious or fatal injury separately; labels are accident counts. The line marks $O/E = 1$. This supporting period comparison contains no traffic adjustment.

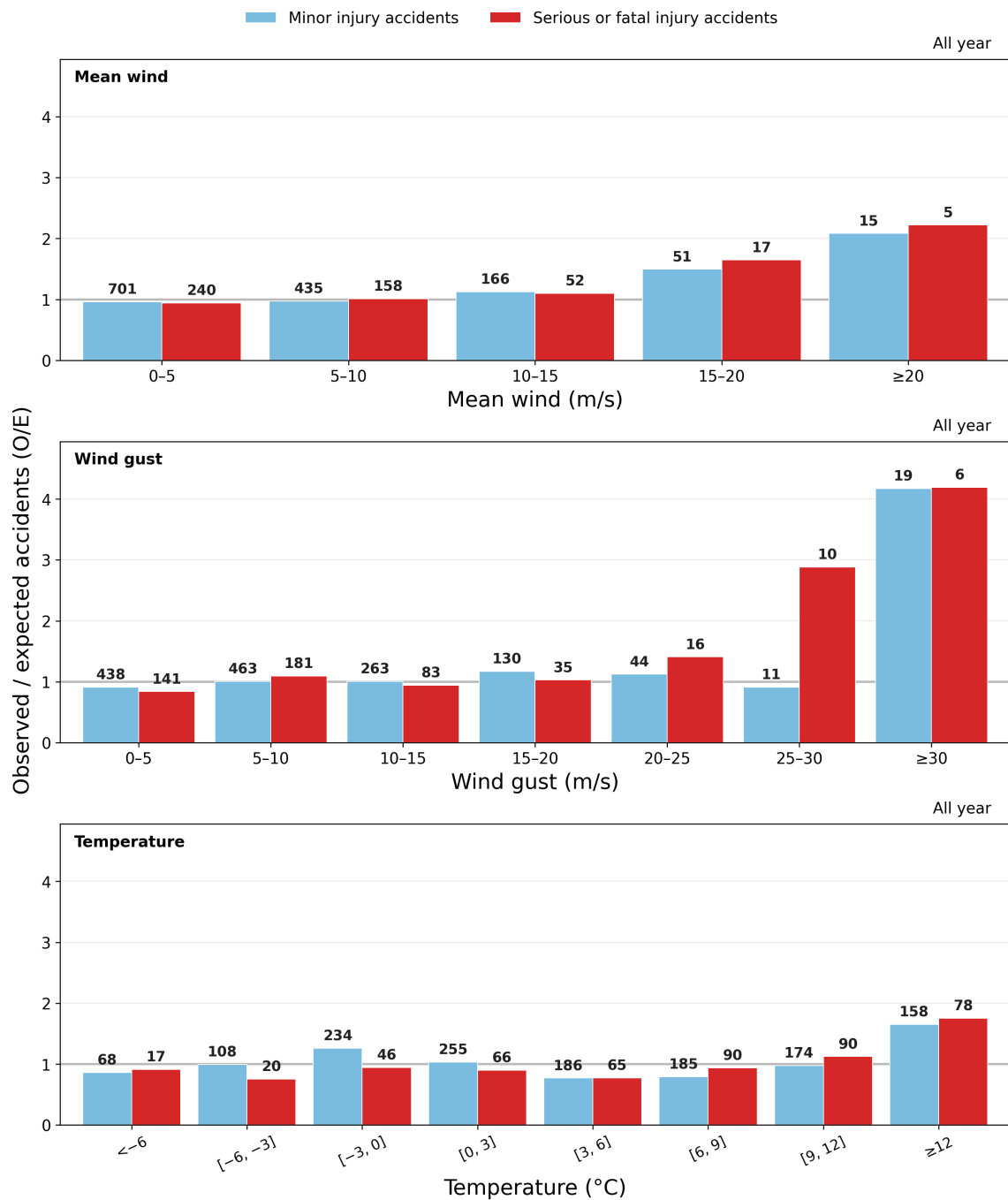


Figure 4.18: Period comparison: All year weather-frequency O/E for rural injury accidents in 2019–2024. Accident counts and station–season background weather are restricted to the same years. Blue and red bars show minor injury and serious or fatal injury separately; labels are accident counts. The line marks $O/E = 1$. This supporting period comparison contains no traffic adjustment.

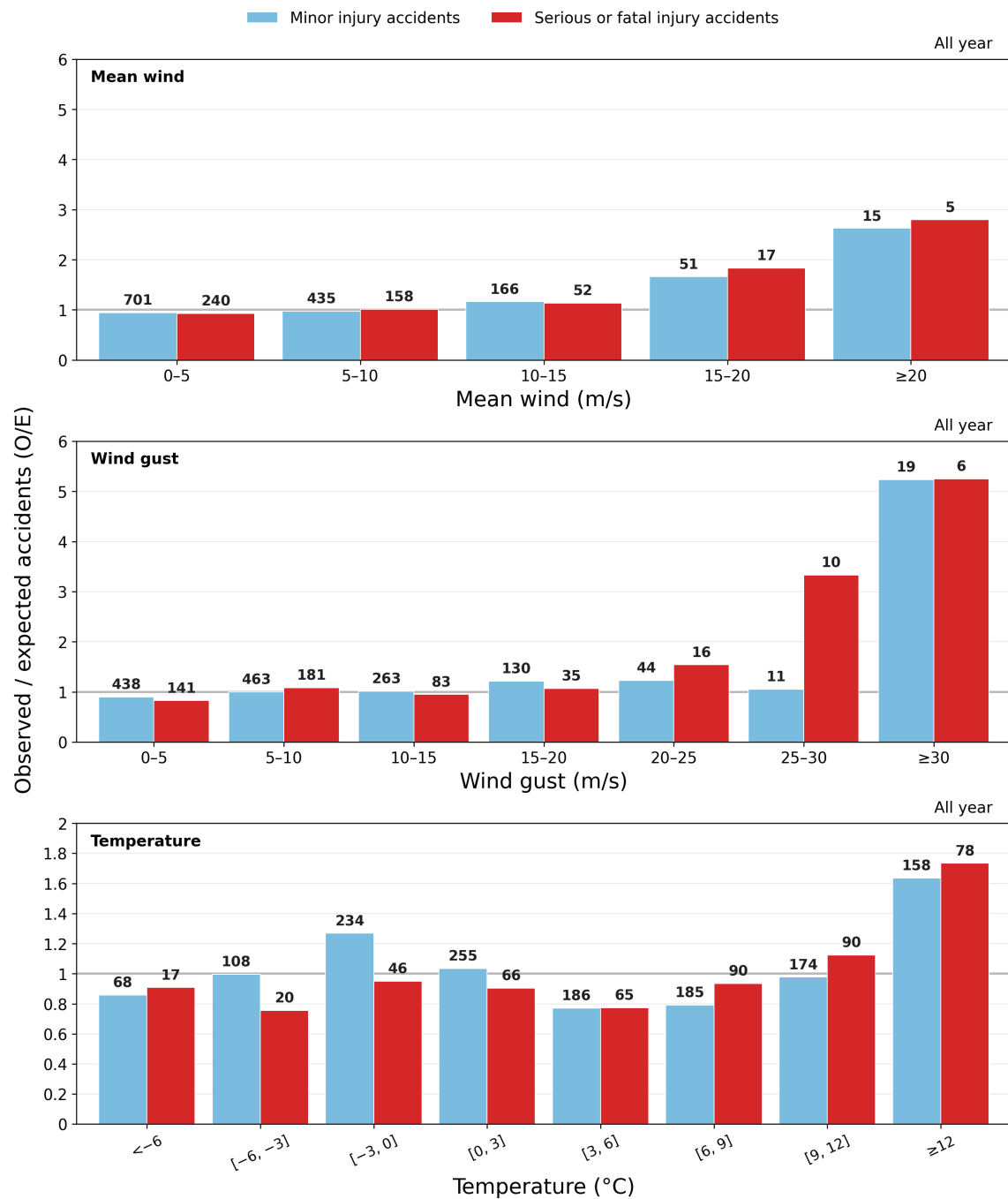


Figure 4.19: Approximately traffic-corrected O/E for qualifying rural injury accidents, 2019–2024, All year. Blue and red bars show minor injury and serious or fatal injury using reweighted expected counts within that period. Labels give observed accidents; the line marks $O/E = 1$. The daily-counter response includes mean wind, gust and temperature, but hourly traffic is still unobserved. This is the correction within the counter era.

4.6.2 Matched-Time and Secondary Weather Checks

The matched-time comparison tests whether the wind pattern persists after controlling for station and recurring calendar time. Table 4.4 shows an upper-wind association, with lower and moderate contrasts near one. This check does not control unusual traffic or road conditions on a particular day.

Table 4.4: Selected matched-time contrasts. Each model uses 6,257 accident strata; the joint wind contrast includes temperature. Intervals are 95% confidence intervals.

Comparison	OR	95% CI
Mean wind: ≥ 15 vs 0–5 m/s	1.61	1.38–1.87
Mean wind, with temperature: ≥ 15 vs 0–5 m/s	1.68	1.44–1.96
Gust: ≥ 30 vs 0–10 m/s	2.74	1.96–3.85
Temperature: $-3-0$ vs $0-3$ °C	1.13	1.02–1.25
Temperature: $3-6$ vs $0-3$ °C	0.59	0.53–0.65
Temperature: ≥ 15 vs $0-3$ °C	0.99	0.80–1.23

The gust model also rises in the upper intervals, whereas the temperature results remain non-monotonic. Including temperature in the mean-wind model changes the upper-wind estimate only slightly. The matched-time temperature result reproduces the dip at moderate temperatures but not the warm-end increase seen in the weather-only analysis, reinforcing the cautious interpretation of temperature.

Daylight and severity provide additional context. Few matched sets change daylight class, making those estimates imprecise. The adjusted severity model provides no clear evidence of greater severity during strong wind once an injury accident has occurred (Appendix Table A.1). A matched-time wind-by-season interaction is detectable, but sparse upper-wind cells do not support ranking individual seasons.

4.6.3 Supporting Traffic Models

The supporting allocated daily-counter model also shows higher rates in strong wind. Its sample and exposure construction differ from the main traffic-rate analysis. Appendix Table A.5 checks how recorded zero counter-days affect the estimated traffic response.

5 Discussion

5.1 Interpretation

The analyses consistently place the strongest contrasts in the upper wind intervals. The weather-only analysis establishes over-representation relative to local weather frequency; the approximate correction asks whether reduced traffic explains it; the traffic-based rate analysis relates accidents to estimated distance travelled on counter-linked roads. Their agreement in direction matters more than agreement in magnitude because they measure different quantities.

The approximate correction addresses an important limitation of weather-frequency O/E: traffic volume is not constant across weather conditions. At the selected counters, daily traffic is lower during stronger wind. If that response also applies to the broader rural road network, weighting expected accidents by the observed traffic response makes the upper-wind contrast larger rather than explaining it away. The correction remains approximate, however. It extrapolates a 2019–2024 counter relationship to the 2007–2025 primary population, and it uses daily traffic totals rather than traffic observed in the same 10-minute intervals used to classify accidents. The approximate correction is therefore best interpreted as a sensitivity analysis of the primary O/E rather than as a measured traffic-adjusted accident rate.

The traffic-based rate analysis incorporates day-to-day traffic variation directly in the denominator, but its larger rate ratio requires separate interpretation. The sample covers only roads and years with usable counter information, and the upper wind intervals contain few accidents. In addition, pooled vehicle-kilometre rates can be influenced by which road sections contribute exposure to each wind interval. A diagnostic comparison within counter-section-year-season strata gives smaller contrasts than the pooled traffic-based rate ratios. This suggests that differences in road and exposure composition account for part of the larger pooled contrasts. It does not establish that particular roads are intrinsically more dangerous, and the sparse upper bins leave substantial uncertainty about the exact magnitude of the contrast.

Monthly-frequency allocation is the preferred main denominator for the available daily traffic totals. Same-day allocation is shown as a sensitivity because it uses realised daily weather. Agreement in the high-wind pattern supports the direction of the association, while differences in magnitude are expected from the different allocations. Neither method observes hourly traffic.

The supporting analyses are consistent with this interpretation but answer different questions. The matched-time comparison retains an upper-wind association after controlling for station and recurring calendar time. This makes an explanation based solely on regular differences between months, weekdays or times of day less plausible. Gust shows a similar upper-tail pattern, although mean wind and gust are related measurements from the same weather observations. Temperature is less consistent: its O/E pattern is non-monotonic and the matched-time analysis does not reproduce the warm-end increase. The severity model

likewise gives no clear evidence that strong wind is associated with a greater probability of serious or fatal injury once an injury accident has occurred. These results support a distinction between the occurrence of accidents and their severity.

The main wind result is consistent with earlier work comparing accident shares with the time spent in adverse weather (Edwards, 1994) and with engineering research showing mechanisms by which crosswind can impair vehicle stability (Baker, 1986, 1994). The contribution here is not a new physical wind threshold, but a long Icelandic accident series linked to local weather frequency and then examined under several complementary exposure constructions. The non-monotonic temperature findings are also consistent with previous evidence that temperature associations can vary by range and season (Lee et al., 2014; Zhan et al., 2020), while the observed reduction in traffic during strong wind agrees with recorder-based evidence that adverse weather changes travel demand (Sathiaraj et al., 2018).

5.2 Strengths

A principal strength is the broad primary sample: rural injury accidents are linked to quality-controlled 10-minute weather observations, and expected counts are standardised within weather station and study season rather than against a single national weather distribution. Match coverage, observed counts and weather-quality exclusions are reported explicitly, which makes the primary O/E calculation transparent and auditable.

The exposure analyses complement rather than replace this broad comparison. The approximate correction tests how the primary result changes under an empirically estimated traffic response, while the traffic-based rate analysis uses observed daily traffic, road length and all eligible counter-days, including days without an accident. The matched-time and additional traffic models provide supporting checks under different conditioning and exposure assumptions. Keeping these analyses separate makes their assumptions visible instead of combining incompatible data sources into one apparently precise measure. Documented transformations, automated tests and stage-specific audits further support reproducibility.

5.3 Limitations

The main limitation is incomplete traffic exposure. The weather-only analysis uses weather time rather than vehicle-kilometres. The approximate correction introduces traffic only through an approximate response estimated from selected counters in 2019–2024 and extrapolated to the broader 2007–2025 population. The counter data are daily totals, so neither the approximate correction nor the traffic-based rate analysis observes how many vehicles were travelling during the specific 10-minute weather interval in which an accident occurred.

The traffic-based rate analysis is more exposure-based but also more selective. It is limited to rural counter sections with usable traffic, road geometry and weather in 2019–2024. The full observed daily count is allocated to 07:00–24:00 using pooled station-month weather frequencies, an assumption that preserves daily exposure but does not reconstruct hourly traffic. The monthly weather frequencies use the longer 2007–2025 weather record for stability, while the traffic and accidents themselves are restricted to 2019–2024. Few the

traffic-based rate analysis accidents occur in the highest mean-wind and gust intervals, so the corresponding large rate ratios should be interpreted as directional evidence rather than precise estimates. An assigned counter also represents exposure on a road section; it does not establish that the vehicle involved in an accident passed that counter.

Weather measurement introduces a separate source of uncertainty. A station up to 20 km from an accident or counter location is a proxy for conditions on the road, and Icelandic terrain can produce substantial variation over shorter distances. The five-minute matching rule can also select an observation on the adjacent calendar day for accidents close to midnight; this preserves the time tolerance but should be understood as part of the matching definition. The fixed 2020–2024 urban-boundary dataset is applied to the full 2007–2025 accident period, so historical changes in settlement boundaries are not represented.

The supporting models have their own restrictions. The case-crossover design controls recurring calendar time but not unusual traffic, road-surface conditions, visibility or closures on a particular day. More generally, all analyses are observational. Unmeasured road characteristics, vehicle type, speed, visibility, driver behaviour and weather-related road management may contribute to the observed patterns. The study therefore supports associations and exposure contrasts, not causal effects of wind.

5.4 Future Work

The most valuable extension would be verified hourly traffic linked to stable counter coordinates, road-section identifiers, direction information and counter downtime. Hourly exposure would remove the need to distribute daily traffic totals across weather intervals and would allow accident-time weather to be paired with traffic measured at a compatible temporal resolution. A second useful extension would examine baseline accident rates and exposure composition across road sections to determine how much of the difference between pooled traffic-based rates and locally standardised O/E is attributable to road mix. With sufficiently detailed data, vehicle class, road geometry and wind direction relative to the road could also be incorporated.

6 Conclusion

Strong wind is over-represented at rural injury accident times in Iceland relative to how often it occurs locally. At mean wind speeds of at least 20 m/s, the primary observed-to-expected ratio is 2.15. Gust shows a similar pattern, while temperature associations are less consistent.

Accounting approximately for reduced traffic strengthens the wind contrast. The separate traffic-based rate analysis points in the same direction, but covers a smaller, more selective road sample and has few accidents in the highest wind intervals. Road and exposure composition appear to explain part of its larger contrast.

These findings support an association, not a causal effect of wind or a direct national estimate of accident risk per distance travelled. Broader counter coverage and hourly traffic measurements are needed to estimate that risk without relying on within-day exposure assumptions.

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A Supporting Checks

This appendix contains the checks most relevant to interpreting the main results. Additional checks are documented separately.

A.1 Accident Characteristics

Figures A.1 and A.2 complement the hour, season and daylight counts in the data chapter. They describe the selected injury accidents without a traffic-exposure denominator.

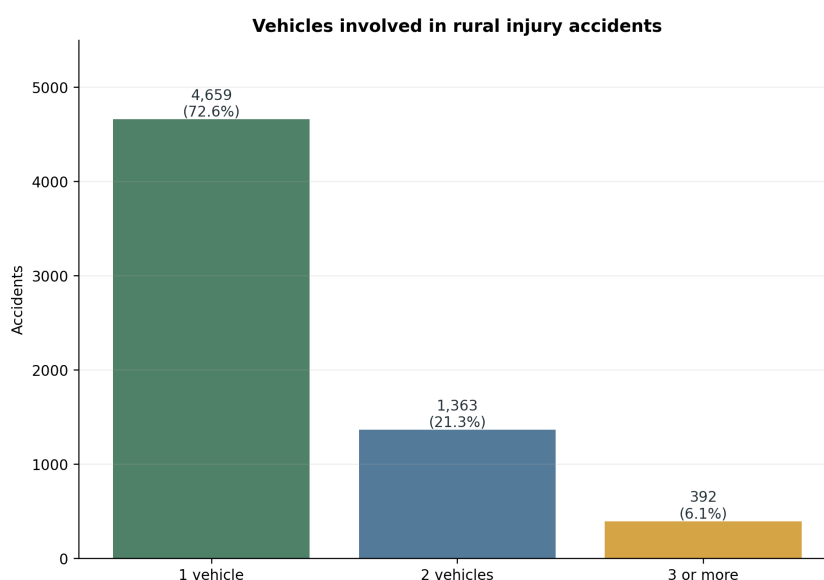


Figure A.1: Number of registered vehicles involved in the 6,414 rural injury accidents. Counts describe the selected sample, not risk by vehicle count.

A.2 Adjusted Severity Model

Table A.1 shows the adjusted comparison of serious or fatal and minor-injury accidents. It describes severity among recorded injury accidents; it does not estimate accident occurrence or risk per vehicle-kilometre.

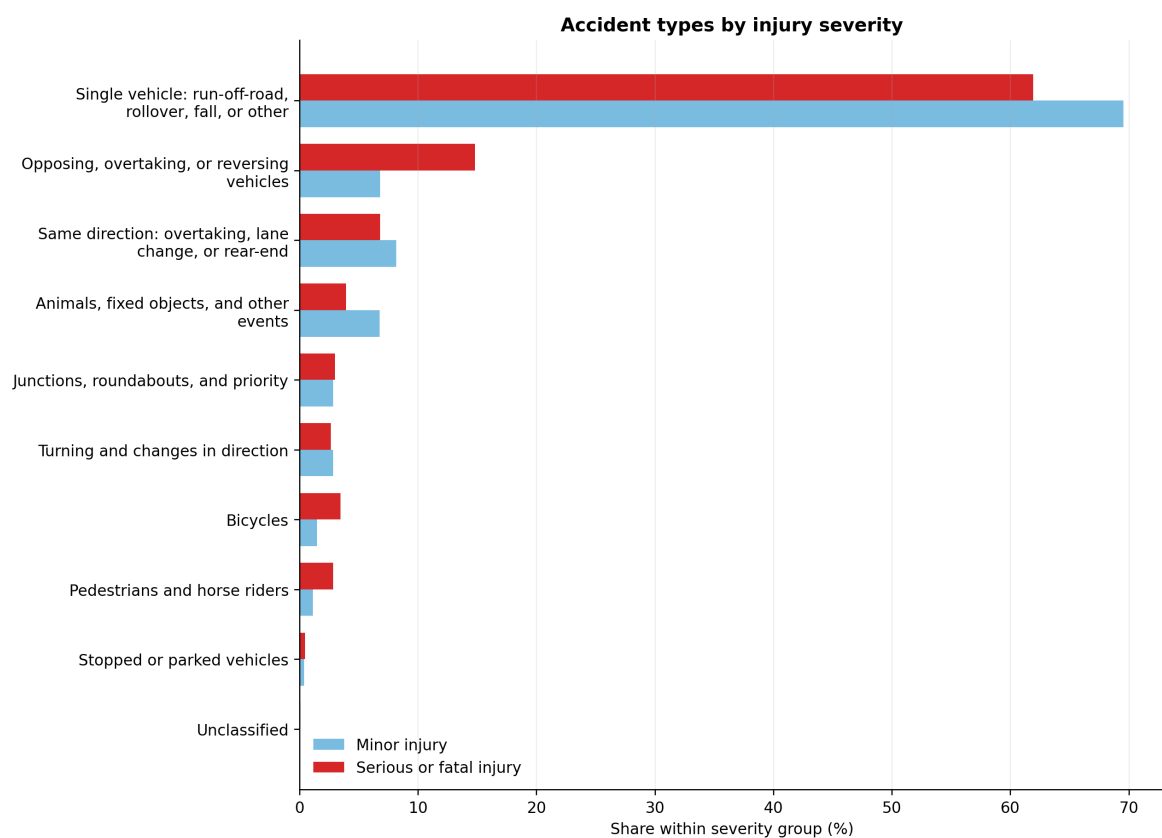


Figure A.2: Accident-type composition for minor injury (blue, code 3) and serious or fatal injury (red, codes 1–2). Percentages sum to 100 within each separate severity group.

Table A.1: Adjusted odds ratios for a serious or fatal outcome among recorded injury accidents. All listed variables are included in the same model.

Variable	Comparison	Adjusted OR (95% CI)
Mean wind	10–15 m/s vs 0–10 m/s	0.99 (0.82–1.21)
Mean wind	≥ 15 m/s vs 0–10 m/s	0.93 (0.69–1.26)
Temperature	< -6 °C vs 0–3 °C	1.13 (0.79–1.62)
Temperature	-6 – -3 °C vs 0–3 °C	0.85 (0.63–1.15)
Temperature	-3 –0 °C vs 0–3 °C	0.83 (0.66–1.04)
Temperature	3–6 °C vs 0–3 °C	1.11 (0.88–1.39)
Temperature	6–9 °C vs 0–3 °C	1.41 (1.10–1.79)
Temperature	9–12 °C vs 0–3 °C	1.28 (0.98–1.68)
Temperature	12–15 °C vs 0–3 °C	1.33 (0.98–1.81)
Temperature	≥ 15 °C vs 0–3 °C	1.21 (0.83–1.76)
Daylight	Darkness vs Daylight	0.89 (0.71–1.12)
Daylight	Civil twilight vs Daylight	1.09 (0.87–1.36)
Time of day	00–06 vs 10–16	0.86 (0.64–1.16)
Time of day	06–10 vs 10–16	0.65 (0.52–0.82)
Time of day	16–20 vs 10–16	0.98 (0.84–1.14)
Time of day	20–24 vs 10–16	0.77 (0.61–0.98)
Season	Winter vs Summer	0.82 (0.64–1.05)
Season	Spring vs Summer	0.96 (0.78–1.18)
Season	Autumn vs Summer	0.83 (0.65–1.05)

A.3 Supporting Allocated Daily-Counter Sample

This section documents the supporting allocated daily-counter model, not the 694-accident traffic-based rate analysis. The tables show how the 2019–2024 rural injury sample narrows under the linkage used by that supporting model. The 767 records at the last broad selection step include five records that do not enter its final allocated-rate model, which retains 762 accidents.

Table A.2 traces the linkage exclusions for this supporting sample.

Table A.2: Selection of accidents for the supporting allocated daily-counter model.

Selection step	Accidents	Share of 2019–2024 sample
Rural injury accidents in 2019–2024	1,863	100.0%
Exact road-section/year counter candi-date	864	46.4%
Assigned counter within 20 km	863	46.3%
Positive count and valid full-day wind	767	41.2%

Table A.3 compares retained accidents with the broader sample and excluded groups by severity, vehicles and season, making selection differences visible.

Table A.3: Characteristics of accidents retained and excluded by the linkage used for the supporting allocated daily-counter model.

Characteristic	All	No link	Linked, excluded	Retained
Accidents	1,863	999	102	762
Share of all accidents	100.0%	53.6%	5.5%	40.9%
Distinct road sections	563	374	56	189
Serious or fatal	25.6%	29.4%	18.6%	21.5%
One vehicle	67.7%	75.2%	54.9%	59.7%
Winter	32.6%	28.0%	40.2%	37.5%
Spring	13.0%	11.7%	8.8%	15.4%
Summer	36.0%	41.8%	33.3%	28.6%
Autumn	18.4%	18.4%	17.6%	18.5%

The geographic selectivity of this traffic-linked sample is documented in Figure A.3.

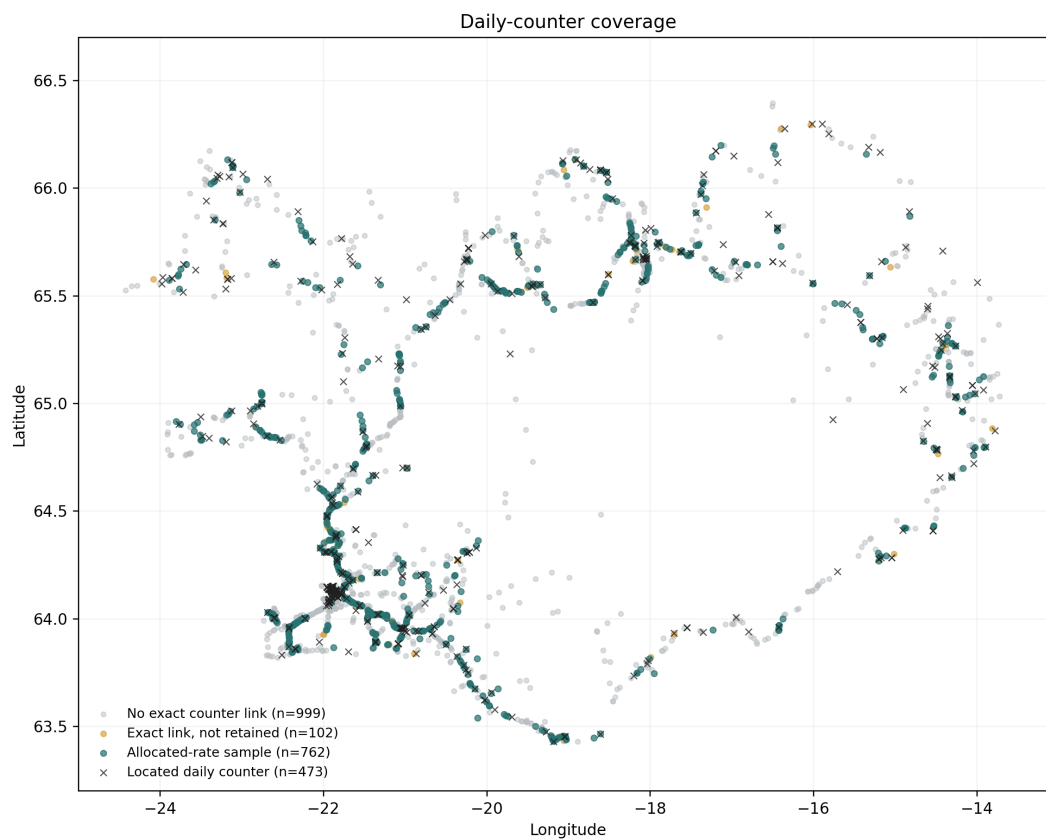


Figure A.3: Geographic coverage of located daily counters and rural injury accidents retained or excluded by the daily-counter linkage, 2019–2024.

A.4 Traffic Sensitivity Checks

Table A.4 shows how the supporting allocated daily-counter result changes under 5, 10, and 20 km limits. It is retained to make the spatial selection rule explicit.

Table A.4: Upper-bin estimate from the supporting allocated daily-counter model under three assignment distances.

Maximum distance	Included accidents	Upper-bin accidents	Rate ratio	95% CI
5 km	585	12	2.91	1.58–5.37
10 km	750	14	2.78	1.58–4.88
20 km	767	14	2.73	1.55–4.78

Table A.5 checks the effect of excluding recorded zero counter-days from the daily traffic-response estimate.

Table A.5: Effect of excluding zero counter-days on the daily traffic response.

Check	Data included	Estimate (95% interval)	Records
Daily traffic, 20–25 m/s	Retain zero counter-days	68.94% (62.33–74.09)	1,175 days
Daily traffic, 20–25 m/s	Exclude zero counter-days	69.29% (62.70–74.52)	1,090 days